Comparative Study of MCMC Sampling Strategies for the 2D Ising Model

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STAT 9250 (Computational Statistics)
Spring 2026

Abstract

The two-dimensional Ising model is one of the few statistical models that admits an exact solution in the infinite-volume limit, making it a controlled testbed for benchmarking MCMC estimators against a known ground truth. We implement single-spin Metropolis–Hastings and the Wolff cluster algorithm from scratch in R with Rcpp inner loops and compare them on $L \in \{8, 16, 32, 64\}$ square lattices across a temperature grid refined around the critical point $T_c = 2/\ln\left(1+\sqrt{2}\right) \approx 2.269$. We quantify mixing through the integrated autocorrelation time (Sokal's adaptive windowing) and effective sample size, and we build confidence intervals via the block bootstrap with block length $\lceil 2\tau_{\text{int}} \rceil$.

At $L = 64$, $T = T_c$, Wolff decorrelates magnetization roughly 79 000 times faster than Metropolis in equivalent-sweep units; the Metropolis dynamic critical exponent is $z_M \approx 2.08$, within 4% of the Nightingale–Blöte literature value 2.17, while Wolff's τ_M saturates at the Sokal floor of 0.5 at every L tested at T_c. The ratio of block-bootstrap to naive i.i.d. SE tracks the theoretical inflation factor $\sqrt{2\tau_E}$ across the temperature grid, with median 2.31× for Wolff and 2.05× for Metropolis at $L = 32$. Applied to the susceptibility-peak extrapolation $T_{\text{peak}}(L) = T_c + a/L$ on $L \geq 16$, the procedure produces a point estimate $\widehat{T_c} = 2.26915$ that agrees with the exact Onsager value to four decimal places, with a 95% confidence interval of $[2.25956, 2.27629]$ covering the exact T_c.

Keywords: MCMC; 2D Ising model; Wolff cluster algorithm; critical slowing down; integrated autocorrelation time; block bootstrap.

1 Introduction

The two-dimensional Ising model occupies an unusual position in computational statistics: it is one of the few non-trivial systems with a closed-form analytical solution, and simultaneously one of the canonical examples where standard MCMC samplers fail. The exact infinite-volume critical temperature $T_c = 2/\ln\left(1+\sqrt{2}\right) \approx 2.2692$ and the analytical forms of the energy and specific heat due to Onsager [1] and the spontaneous magnetization due to Yang [2] provide ground truth for any estimator. At the same time, near T_c most of the textbook single-site samplers degrades catastrophically through critical slowing down, which means the integrated autocorrelation time τ_{int} (the average number of MCMC steps required to produce one effectively independent sample) grows as $\tau_{\text{int}} \sim L^z$ with $z \approx 2.17$ for single-spin Metropolis [3], so the inferential cost of each additional lattice doubling grows by a factor of nearly five. Few problems offer analytical tractability which makes the 2D Ising model a uniquely informative testbed for comparing MCMC strategies under criticality.

We use this testbed to conduct a focused three-part comparison. Two samplers are implemented from scratch: single-spin Metropolis–Hastings and the Wolff cluster algorithm [4] and benchmarked against each other and against the analytic ground truth using a diagnostic pipeline built around Sokal's adaptive-window

estimator for τ_{int}. The block bootstrap [5, 6] is then layered on top of the MCMC output to produce standard errors and confidence intervals that respect the autocorrelation structure of the chain, with explicit calibration against the naive i.i.d. alternative.

The main narrative of the paper is that the cluster algorithm's mixing advantage near criticality is not just a sampling acceleration: it is the enabler of every downstream inference in the study. Wolff's near-iid sampling at and below T_c produces effective sample sizes orders of magnitude larger than Metropolis at the same chain length, and this gap propagates directly into the width of the bootstrap confidence intervals: at $L = 32$, $T = T_c$ with $n = 10\,000$ samples, Metropolis bootstrap standard errors are 12 to 19× wider than Wolff's for magnetization-derived observables. The block bootstrap correctly diagnoses this inflation, and the ratio of block-bootstrap SE to naive i.i.d. SE tracks the theoretical $\sqrt{2\tau_{\text{int}}}$ prediction across the entire temperature grid for both samplers. The naive bootstrap, by contrast, is systematically anti-conservative precisely near T_c where uncertainty quantification matters most. Finally, the cluster algorithm's CPU efficiency is what makes a dense 34-point temperature sweep at $L = 64$ computationally feasible, a sweep that would be prohibitive with Metropolis at the chain lengths required for comparable precision, and that sweep, combined with a finite-size scaling extrapolation of the susceptibility peak, recovers the exact Onsager T_c to four decimal places with valid 95% coverage.

Paper organization. Section 2 surveys the relevant literature on exact solutions of the Ising model, single-site and cluster MCMC samplers, autocorrelation diagnostics, and the block bootstrap for dependent data. Section 3 defines the model, observables, and exact ground-truth formulas used throughout as reference. Section 4 describes the samplers, mixing diagnostics, block bootstrap, T_c estimator, and simulation design. Section 5 presents the empirical findings, organized as a flow from chain diagnostics to the output of physical quantities: sampler mixing, dynamic critical exponents, validation against ground truth, phase-transition observables with bootstrap uncertainty, bootstrap calibration, head-to-head bootstrap comparison, and the T_c estimate. Section 6 interprets these findings, discusses limitations, and outlines directions for future work, and Section 7 provided the concluding remark.

2 Literature Review

The 2D Ising model sits at a three-way intersection of statistical mechanics, Markov chain Monte Carlo theory, and time-series statistics. In this section we summarize the relevant literature involded in these areas.

Exact solutions and finite-size scaling. Onsager's 1944 solution [1] is widely regarded as one of the most important results in 20th-century statistical physics. It derived the exact critical temperature, the energy and specific heat in closed form, and established that 2D systems can exhibit genuine continuous phase transitions. Yang [2] closed the remaining gap with the spontaneous magnetization. The infinite-volume exactness, however, is also a limitation as real simulations are always finite, and the divergent quantities (c and χ at T_c) require finite-size scaling theory [7] to connect simulation output to the analytical predictions. Modern treatments such as Newman and Barkema [8] and Sokal [9] provide pedagogical access to both the exact results and the finite size scaling framework, and serve as the standard references for Monte Carlo simulation of the model.

Single-site sampling and its discontents. The Metropolis–Hastings algorithm [10] confirms the recognition that single-site updates suffer slowdown near phase transitions. The phenomenon, now called critical slowing down, was characterized through the dynamic critical exponent z, defined by $\tau_{\text{int}} \sim L^z$ at criticality. For 2D Ising under single-site Metropolis, the consensus value $z = 2.1665 \pm 0.0012$ is due to Nightingale and Blöte [3], obtained via transfer-matrix extrapolation rather than direct Monte Carlo which is a methodologically important

point, since direct fits from MCMC chains are themselves plagued by the slow convergence they purport to measure. The Nightingale–Blöte value remains the benchmark against which more recent algorithmic claims are tested.

Cluster algorithms. The first major breakthrough against critical slowing down came from Swendsen and Wang [11], who exploited the Fortuin–Kasteleyn correspondence [12] between the Ising model and a random-cluster representation to design updates that flip entire correlated clusters at once. Wolff [4] introduced a simpler single-cluster variant that is now the more common reference implementation; Wolff also reported a much smaller dynamic exponent $z \approx 0.25$ for the magnetization channel, amounting to a near-elimination of critical slowing down for that observable. The literature on whether z^{Wolff} is exactly zero or merely very small remains active [9], and the question of whether Swendsen–Wang or Wolff is faster in practice depends on observable, lattice size, and dimensionality [13]. A common gap in this literature is that autocorrelation speedups are typically reported as ratios of τ_{int} without translation into the inferential currency of confidence interval width, leaving practitioners to perform their own bookkeeping when applying these algorithms in statistical practice.

Bootstrap for autocorrelated chains. Efron's i.i.d. bootstrap [14] destroys the autocorrelation structure of an MCMC chain and underestimates SEs by $\sqrt{2\tau_{\text{int}}}$ in expectation. The block bootstrap was developed in two stages to address this: Carlstein [6] introduced non-overlapping block resampling, partitioning the chain into disjoint blocks of length ℓ, and Künsch [5] later proposed the moving block bootstrap using all $n - \ell + 1$ overlapping blocks; the two variants are asymptotically equivalent under mild mixing conditions. The standard prescription [5] is $\ell \propto \tau_{\text{int}}$; the constant of proportionality is the central tuning question, with theoretical work showing that the MSE-optimal rate for variance estimation is $\ell \sim n^{1/3}$ [15].

Synthesis. The four threads above are well-developed individually but rarely combined in a single statistical inferential study. The physics literature emphasizes algorithmic speedups in autocorrelation units; the statistics literature emphasizes block-bootstrap consistency on generic time series. Standard treatments of MCMC for the Ising model (such as [8]) discuss autocorrelation but not bootstrap uncertainity quantification. Standard treatments of the block bootstrap (such as [16]) discuss general time series but rarely the specific challenges of critical slowing down. This project builds upon at that intersection.

3 Model and Exact Ground Truth

3.1 Hamiltonian and observables

We study the ferromagnetic Ising model on a two-dimensional square lattice $\Lambda = \{1, \ldots, L\}^2$ with periodic boundary conditions, where L is the linear lattice size and $N = L^2$ is the total number of sites. A configuration is an assignment of a spin $s_i \in \{-1, +1\}$ to each site $i \in \Lambda$, collected into the vector $\mathbf{s} = (s_1, \ldots, s_N)$. The energy of a configuration, in the absence of an external magnetic field, is given by the Ising Hamiltonian [8, Eq. (3.1)]

$$H(\mathbf{s}) \;=\; -J \sum_{\langle i,j \rangle} s_i s_j, \tag{1}$$

where the sum runs over nearest-neighbor pairs $\langle i, j \rangle$ (each unordered pair counted once) and the coupling constant $J > 0$ sets the strength of the ferromagnetic interaction. Throughout we fix $J = 1$ and Boltzmann's constant $k_B = 1$, so the inverse temperature is $\beta = 1/T$ and the Boltzmann probability of a configuration at temperature T is $\pi_T(\mathbf{s}) \propto e^{-\beta H(\mathbf{s})}$ [8, Eq. (1.5)].

The total magnetization of a configuration is $M(\mathbf{s}) = \sum_{i\in\Lambda} s_i$ [8, Eq. (1.33)], and expectations $\langle \cdot \rangle$ are taken with respect to π_T. The four per-spin thermodynamic observables we estimate throughout this paper are the energy, the absolute magnetization, the specific heat, and the magnetic susceptibility. The equations are as follows:

$$u(T) = \langle H \rangle / N, \tag{2}$$

$$|m|(T) = \langle |M| \rangle / N, \tag{3}$$

$$c(T) = \frac{\left\langle H^2 \right\rangle - \langle H \rangle^2}{NT^2}, \tag{4}$$

$$\chi(T) = \frac{\left\langle M^2 \right\rangle - \langle |M| \rangle^2}{NT}. \tag{5}$$

Equation (2) is the mean energy per spin; (3) follows the convention [8, §3.7.1, Eq. (1.34)] of averaging $|M|$ rather than M to obtain a meaningful finite-lattice order parameter below T_c, as discussed below. Equations (4) and (5) are the fluctuation-based expressions for the specific heat and susceptibility, which follow from the fluctuation–dissipation theorem applied to H and M respectively [8, Eqs. (1.36),(1.37),(3.15),(3.16)].

We use $\langle |M| \rangle^2$ in Equation (5) rather than $\langle M \rangle^2$ (as appears in Eq. (1.36) of Ref. [8]) because on a finite lattice in zero external field the magnetization distribution is symmetric under $M \to -M$. Over a long simulation the chain visits both symmetry-related ordered phases below T_c, so $\langle M \rangle \to 0$ and the naive variance $\left\langle M^2 \right\rangle - \langle M \rangle^2 = \left\langle M^2 \right\rangle$ picks up contributions from both phases, inflating χ by their inter-phase separation. Following the convention in [8, §3.7.1], which uses $\langle |m| \rangle$ rather than $\langle m \rangle$ as the order parameter for exactly this reason, we likewise substitute $\langle |M| \rangle^2$ into the variance expression for χ so that the two are defined consistently.

The expectations defining Equations (2) to (5) are sums over the configuration space $\{-1, +1\}^N$, whose cardinality 2^N grows astronomically with the lattice size: even the smallest grid we study, $L = 8$ ($N = 64$), has $2^{64} \approx 1.8 \times 10^{19}$ configurations, and the largest, $L = 64$ ($N = 4096$), has 2^{4096} — a number with more than 1200 decimal digits. Direct enumeration is therefore infeasible at every lattice size of interest, and the high-dimensional integration must instead be approximated by sampling configurations according to the Boltzmann distribution $\pi_T(\mathbf{s}) \propto e^{-\beta H(\mathbf{s})}$ and replacing expectations with empirical averages. Markov chain Monte Carlo (MCMC) is the standard tool for generating such samples; the remainder of this paper compares two specific MCMC strategies for this purpose. Closed-form expressions for $u(T)$, $c(T)$, and $|m|(T)$ in the infinite-volume limit are nevertheless available for the 2D Ising model and are reviewed in Section 3.2 below; these provide the analytical ground truth against which our MCMC estimates are validated.

Remark 1 (A known finite-lattice bias)**.** The $\langle |M| \rangle$ convention in Equation (3) introduces a positive bias of order $1/\sqrt{N}$ above T_c, where the true infinite-volume magnetization is zero: with $M \sim \mathcal{N}(0, \sigma^2)$ in the disordered phase, $\langle |M| \rangle = \sigma\sqrt{2/\pi} > 0$ despite $\langle M \rangle = 0$. This is a known artifact of finite-lattice Monte Carlo [8, §3.7.1 and footnote 16] and appears in our results as a small but visible offset between our MCMC estimates and Yang's curve for $|m|(T > T_c)$ at $L = 8, 16$ (see Section 5.3 and Figure 5). In Figure 5 in the left magnetization panel (full range), at region $T > 2.27$, dots for $L = 8, 16$ clearly sit above zero.

3.2 Onsager and Yang exact solutions

The model undergoes a continuous phase transition at the critical temperature [1, Eq. (39a)]

$$T_c = \frac{2J}{\ln\left(1 + \sqrt{2}\right)} \approx 2.269185\, J, \tag{6}$$

obtained by setting $H = H^* = \frac{1}{2} \log \cot(\pi/8)$ in Onsager's duality relation (his Eq. (17)), where $H = \beta J$. For $T < T_c$ the system exhibits spontaneous symmetry breaking and develops a non-zero magnetization $|m| > 0$;

for $T > T_c$ it is in a disordered (paramagnetic) phase with $|m| \to 0$ as $L \to \infty$. At $T = T_c$ the correlation length diverges and the specific heat diverges logarithmically, producing a second-order phase transition. Physically, T_c is the temperature at which a ferromagnet loses its permanent magnetization: below T_c thermal fluctuations are too weak to overcome the alignment-favoring nearest-neighbor coupling and the spins lock into a globally ordered state, while above T_c thermal energy dominates and the spins fluctuate independently. Right at T_c the two effects balance exactly, producing fluctuations on every length scale (the divergent correlation length) this explains why MCMC samplers built around local moves slow down catastrophically in this regime.

Energy and specific heat. Onsager's exact infinite-lattice energy and specific heat per spin are expressed in terms of the complete elliptic integrals of the first and second kind, defined as

$$K(k) \;=\; \int_0^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{1-k^2\sin^2\theta}}, \qquad E(k) \;=\; \int_0^{\pi/2} \sqrt{1-k^2\sin^2\theta}\,\mathrm{d}\theta, \tag{7}$$

with modulus k_1 and signed complementary modulus k_1'' defined by [1, Eq. (114)]

$$k_1(T) \;=\; \frac{2\sinh(2\beta J)}{\cosh^2(2\beta J)}, \qquad k_1''(T) \;=\; 2\tanh^2(2\beta J) - 1. \tag{8}$$

Here, k_1'' differs from the standard complementary modulus $k_1' = \sqrt{1-k_1^2}$ by a sign below T_c; we follow Onsager's convention [1] throughout. In these terms the energy per spin is [1, Eq. (116)]

$$u(T) \;=\; -J\coth(2\beta J)\left[1 + \tfrac{2}{\pi}\big(2\tanh^2(2\beta J) - 1\big)\,K\big(k_1(T)\big)\right], \tag{9}$$

and the specific heat per spin is [1, Eq. (117)]

$$\begin{aligned} c(T) \;&=\; \tfrac{2}{\pi}\big(\beta J\coth(2\beta J)\big)^2 \\ &\quad\times\Big[2K(k_1) - 2E(k_1) - (1-k_1'')\big(\tfrac{\pi}{2} + k_1''\,K(k_1)\big)\Big]. \end{aligned} \tag{10}$$

As $T \to T_c$, the modulus $k_1 \to 1$ and $K(k_1)$ diverges logarithmically as $K(k_1) \sim \log\big(2^{5/2}/|H - H_c|\big)$ [1, Eq. (120)], which produces the well-known log singularity of $c(T)$ at the critical point. The critical values of the thermodynamic functions are summarized in Onsager's critical-data tabulation [1, p. 141, Eq. (121)]; most relevant for our validation is the energy per spin at the critical point,

$$u(T_c) \;=\; -\sqrt{2}\,J \;\approx\; -1.41421356\,J, \tag{11}$$

which we use as the exact ground-truth reference when comparing our MCMC estimators at $T = T_c$ in Section 5.3.

Spontaneous magnetization. Yang's exact infinite-lattice spontaneous magnetization [2], reduced to its standard compact form, is

$$|m|(T) \;=\; \begin{cases} \big[1 - \sinh^{-4}(2\beta J)\big]^{1/8} & \text{if } T < T_c, \\ 0 & \text{if } T \geq T_c. \end{cases} \tag{12}$$

Susceptibility. No closed-form expression for the infinite-lattice susceptibility $\chi(T)$ is known. However, finite-size scaling theory [7] combined with the critical exponents of the 2D Ising universality class ($\gamma = 7/4$, $\nu = 1$, derivable from Onsager [1]) predicts the scaling relation

$$\chi(T_c; L) \sim L^{\gamma/\nu}, \qquad \gamma/\nu = 7/4, \tag{13}$$

where γ and ν are respectively the susceptibility and correlation-length critical exponents. We use Equation (13) as a validation target for the MCMC estimates of χ at T_c across our four lattice sizes.

Figure 1 shows Equations (9), (10) and (12) on the temperature range used throughout this paper.

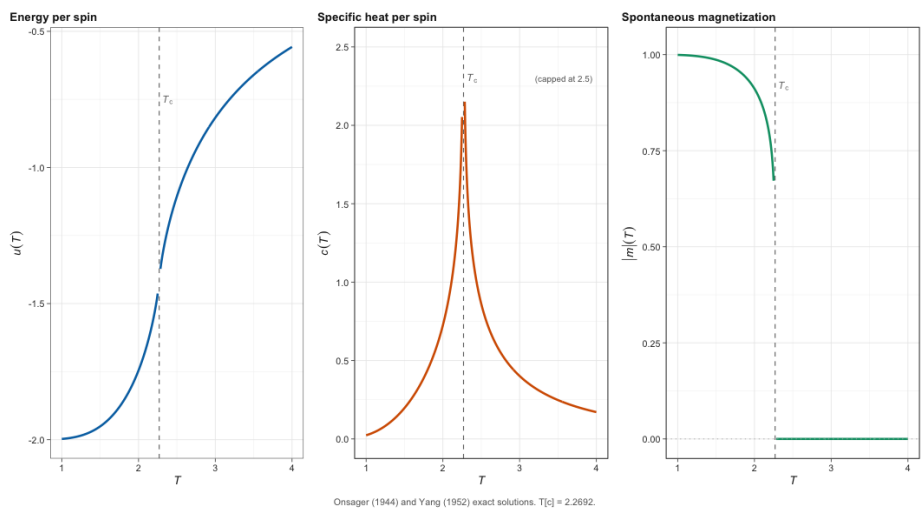


Figure 1. Onsager and Yang exact ground-truth curves used as reference throughout this paper. The complete elliptic integral $K(k_1)$ diverges as $k_1 \to 1$, which occurs at T_c; in numerical evaluation we guard against this by returning $K = \infty$ whenever $|k_1| > 1 - 10^{-10}$, preventing quadrature failure in a small neighborhood of the critical point. For energy per spin, for consistency we omit the curve near critical temperature although we have continuity there.

4 Methods

4.1 Samplers

We compare two Markov chain algorithms that sample from the Boltzmann distribution $\pi_T(\mathbf{s}) \propto e^{-\beta H(\mathbf{s})}$ using different move sets: single-spin flips (Metropolis) and collective cluster flips (Wolff). Both are implemented from scratch in two versions — a transparent pure-R reference that mirrors the pseudocode exactly, and an Rcpp [17] inner-loop version for actual runs.

4.1.1 Single-spin Metropolis–Hastings

The Metropolis algorithm [10], is adapted for The Ising model where each elementary move proposes a single-spin flip, and the acceptance rule follows the standard Metropolis prescription with ΔE computed locally from the four nearest neighbors of the chosen site [8, Eq. (3.10)].

One Metropolis step proceeds as follows:

1. Select a site $i \in \Lambda$ uniformly at random.

2. Compute the local energy change that would result from flipping its spin,

$$\Delta E \;=\; 2Js_i \sum_{j\in\mathcal{N}(i)} s_j, \tag{14}$$

where $\mathcal{N}(i)$ denotes the four nearest neighbors of i under periodic boundary conditions.

3. Accept the flip with probability

$$p_{\text{acc}} \;=\; \begin{cases} 1 & \text{if } \Delta E \le 0, \\ e^{-\beta\Delta E} & \text{if } \Delta E > 0, \end{cases}$$

following [8, Eq. (3.7)]: if $\Delta E \le 0$ the flip is deterministic; otherwise draw $u \sim \text{Uniform}(0,1)$ and accept if $u < e^{-\beta\Delta E}$.

4. If accepted, flip the spin: $s_i \to -s_i$.

The chain records the current configuration after every step regardless of whether the proposed flip was accepted. This preserves balance and canonical distribution can be recovered in the long-run limit, as can be seen in [10, §II].

One Monte Carlo sweep consists of $N = L^2$ attempted flips with sites drawn independently with replacement. We report all autocorrelation and timing results in units of sweeps, so that the cost of one "sweep" is comparable across lattice sizes.

Implementation detail: the 5-entry exponential table. For the 2D Ising model on a square lattice, the neighbor sum $\sum_{j\in\mathcal{N}(i)} s_j \in \{-4,-2,0,+2,+4\}$, so ΔE takes only five values: $\{-8J,-4J,0,+4J,+8J\}$. Our inner loop precomputes a length-5 lookup table of $e^{-\beta\Delta E}$ values at the start of each run and indexes into it in $O(1)$ time during the accept/reject decision. This eliminates the need to compute the exponential in the inner loop entirely and accounts for a large fraction of the Rcpp speedup over pure R (the pure-R version recomputes the exponential per step for clarity). The full pseudocode is given below.

Algorithm 1: Single-spin Metropolis sweep for the 2D Ising model

Input: spin configuration **s**, temperature T, lattice linear size L
Output: updated **s** after one sweep

1. Precompute lookup table: $\text{tab}[k] \leftarrow \exp\big(-(-8+4k)/T\big)$ for $k = 0,1,\ldots,4$ # indexed by $(\Delta E+8)/4$
2. **for** step = 1 **to** L^2 **do**
   (a) draw $i, j \sim \text{Uniform}\{1,\ldots,L\}$
   (b) compute $S \leftarrow s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1}$ # indices mod L
   (c) $\Delta E \leftarrow 2Js_{i,j}S$
   (d) **if** $\Delta E \le 0$ **or** $\text{Uniform}(0,1) < \text{tab}[(\Delta E+8)/4]$ **then** $s_{i,j} \leftarrow -s_{i,j}$
3. **return s**

4.1.2 Wolff cluster algorithm

The single-spin Metropolis chain suffers from critical slowing down near T_c since the integrated autocorrelation time for the magnetization grows as $\tau_M \sim L^{z_M}$ with $z_M = 2.1665 \pm 0.0012$ [3], as reported in [8, Eq. (4.7) and Fig. 4.2]. In our own measurements at $L = 64, T = T_c$, the Metropolis chain produces only $\text{ESS}_M \approx 11$ effectively independent magnetization samples in a run of $n = 320\,000$ sweeps (Table 1). Wolff's collective-update algorithm [4] dissolves this slowdown by flipping an entire correlated *cluster* of aligned spins at once, with the cluster construction chosen so that every proposed move is accepted [8, Eq. (4.13)].

Let $\sigma \in \{-1, +1\}$ denote the value of a seed spin chosen uniformly at random. A cluster is grown from this seed by examining each nearest neighbor of every spin added to the cluster: if the neighbor is aligned with σ and not already in the cluster, the connecting bond is activated (and the neighbor added) with probability

$$p_{\text{add}} \;=\; 1 - e^{-2\beta J}. \tag{15}$$

A neighbor of the opposite sign is never added, and the growth terminates when no spin in the cluster has a neighbor left to test. All spins in the resulting cluster are then flipped simultaneously [4].

We now show that Equation (15) is the unique bond probability that makes every cluster flip acceptable, independent of temperature and configuration. Let μ be a configuration containing a cluster C of like spins, and let ν be the configuration with C flipped. Define

- m = number of boundary bonds in μ (cluster spin aligned with same-sign neighbor outside C, contributing energy $-J$);
- n = number of boundary bonds in ν (cluster spin aligned with same-sign neighbor outside C after the flip, contributing $-J$).

The bonds inside C are unchanged by the flip; only the $m + n$ boundary bonds change sign. Hence the energy difference is

$$E_\nu - E_\mu \;=\; 2J(m - n). \tag{16}$$

The probability that a particular cluster C is built from a given seed is the product of $(1 - p_{\text{add}})$ (the probability that a same-sign boundary bond is tested but not added to the cluster) over the m boundary bonds in μ that were tested and rejected. The reverse move (in ν) builds the same cluster by testing and rejecting n boundary bonds. The ratio of selection probabilities $g(\cdot)$, where $g(\mu \to \nu)$ denotes the probability that the algorithm proposes the move $\mu \to \nu$, is therefore

$$\frac{g(\mu \to \nu)}{g(\nu \to \mu)} \;=\; \frac{(1 - p_{\text{add}})^m}{(1 - p_{\text{add}})^n} \;=\; (1 - p_{\text{add}})^{m-n}. \tag{17}$$

Detailed balance requires $g(\mu \to \nu)\, A(\mu \to \nu) \,/\, g(\nu \to \mu)\, A(\nu \to \mu) = e^{-\beta(E_\nu - E_\mu)}$, where $A(\mu \to \nu)$ is the probability of accepting the proposed move. Substituting (16), (17), and (15) gives

$$\underbrace{(1 - p_{\text{add}})^{m-n}}_{\text{selection ratio}} \cdot \underbrace{\frac{A(\mu \to \nu)}{A(\nu \to \mu)}}_{\text{acceptance ratio}} \;=\; e^{-2\beta J(m-n)}, \tag{18}$$

and with $p_{\text{add}} = 1 - e^{-2\beta J}$ the selection ratio becomes $e^{-2\beta J(m-n)}$, which exactly cancels the Boltzmann factor on the right-hand side. Detailed balance is therefore satisfied with

$$A(\mu \to \nu) \;=\; A(\nu \to \mu) \;=\; 1. \tag{19}$$

Every cluster flip is accepted and no explicit accept/reject step is required. Ergodicity follows because any configuration can be reached from any other through a finite sequence of single-spin clusters [8, § 4.2].

The Wolff algorithm is fully specified by which boundary bonds are tested and with what probability, not by the order in which they are tested: a stack, a queue, or any other traversal strategy produces the same distribution over clusters, as long as every neighbor of every in-cluster spin is eventually tested at least once. Our implementation uses a stack (depth-first traversal): newly added spins are pushed onto the stack, and the next spin to examine is popped from the top. A boolean in-cluster flag array prevents any spin from being added more than once. Both the stack buffer and the flag array are $O(N)$ and pre-allocated once per run. The full pseudocode is as follows:

Algorithm 2: Single Wolff cluster update for the 2D Ising model

Input: spin configuration $\mathbf{s}$, temperature T, lattice linear size L
Output: updated $\mathbf{s}$ after one cluster flip, and the cluster size $|C|$

1. $p_{\text{add}} \leftarrow 1 - \exp(-2/T)$ # $J = k_B = 1$
2. draw seed $(i_0, j_0) \sim \text{Uniform}\{1, \ldots, L\}^2$; let $\sigma \leftarrow s_{i_0,j_0}$
3. initialize stack $\mathcal{S} \leftarrow \{(i_0, j_0)\}$, in-cluster flag array $C \leftarrow \mathbf{0}$, $C_{i_0,j_0} \leftarrow 1$, $|C| \leftarrow 1$
4. **while** $\mathcal{S} \neq \emptyset$ **do**
   (a) pop (i, j) from top of $\mathcal{S}$
   (b) **for each** neighbor (i', j') of (i, j) on the torus **do**
      i. **if** $C_{i',j'} = 1$ **or** $s_{i',j'} \neq \sigma$ **then** continue
      ii. **if** $\text{Uniform}(0, 1) < p_{\text{add}}$ **then** $C_{i',j'} \leftarrow 1$; push (i', j') onto $\mathcal{S}$; $|C| \leftarrow |C| + 1$
5. **for each** site (i, j) with $C_{i,j} = 1$ **do** $s_{i,j} \leftarrow -s_{i,j}$
6. **return** $(\mathbf{s}, |C|)$

Unlike Metropolis (where ΔE is already computed to make the accept/reject decision, so E and M can be updated incrementally at zero additional cost), the Wolff sampler does not compute the cluster's contribution to ΔE during the build. Our implementation therefore recomputes E and M from scratch after each cluster flip using Equation (1) directly; this is $O(N)$ per cluster flip, but amortizes over the $O(L^2)$ spins typically touched per flip near T_c, so the total cost per equivalent sweep is still comparable to Metropolis.

Metropolis and Wolff produce samples at very different rates per algorithmic step. Metropolis attempts one flip per step (and touches at most one spin); Wolff flips a cluster whose size varies from $O(1)$ at high temperature to a significant fraction of N at and below T_c. Reporting τ_{int} in "Wolff cluster flips" versus "Metropolis sweeps" would therefore compare two incommensurate units of Monte Carlo work. We instead report Wolff times in equivalent sweeps where one equivalent sweep equals $N/\langle|C|\rangle$ cluster flips, where $\langle|C|\rangle$ is the mean cluster size over the run. This places both samplers on a common axis of "Monte Carlo work per spin touched", which is the natural unit for mixing comparison.

4.1.3 Implementation

Both samplers are implemented twice. The pure-R reference version of each follows Sections 4.1.1 and 4.1.2 line by line, with no optimization beyond vectorized neighbor lookups. We use this version primarily to double-check the Rcpp output at small L and to serve as a readable specification of the algorithm. The Rcpp [17] versions use flat integer arrays for the lattice and pre-allocated $O(N)$ scratch buffers (the exponential lookup table for Metropolis, and the in-cluster flag array plus stack buffer for Wolff), with neighbor indexing done via integer modulo and no runtime memory allocation inside the inner loop. Both Rcpp versions call `GetRNGstate/PutRNGstate` to share R's RNG state, so every run is exactly reproducible from an R-level `set.seed()` call. Across the 12-cell diagnostic grid $(L, T) \in \{8, 16, 32, 64\} \times \{1.5, T_c, 4.0\}$, the Rcpp Metropolis implementation is a median of 341× faster than pure R, and the Rcpp Wolff implementation is a median of 83× faster. These numbers can vary depending on CPU type and may change at every run.

4.2 Mixing diagnostics

Let $\{X_t\}_{t=0,1,2,\ldots}$ denote the sequence of values of a scalar observable X measured at successive Monte Carlo sweeps of a stationary Markov chain. For our chains, X is either the per-spin energy u or the per-spin magnetization $|m|$, measured after every Metropolis sweep or every Wolff cluster flip. The autocovariance

at lag t is $\gamma(t) = \mathrm{Cov}(X_0, X_t)$ and its normalized form is $\rho(t) = \gamma(t)/\gamma(0)$, with $\rho(0) = 1$ and $\rho(t) \to 0$ as $t \to \infty$ for any ergodic chain.

The integrated autocorrelation time τ_{int} is the factor by which the variance of the sample mean is inflated relative to independent sampling: for a chain of length n with $n \gg \tau_{\mathrm{int}}$,

$$\mathrm{Var}(\bar{X}_n) \;\approx\; \frac{2\tau_{\mathrm{int}}\,\sigma_X^2}{n}, \qquad \sigma_X^2 = \gamma(0), \tag{20}$$

so that the chain contains roughly $n/(2\tau_{\mathrm{int}})$ effectively independent samples [9, Eq. (2.20) and p. 13]. Equivalently, τ_{int} admits the series representation [9, Eq. (2.16)]

$$\tau_{\mathrm{int}} \;=\; \tfrac{1}{2} + \sum_{t=1}^{\infty} \rho(t), \tag{21}$$

in which the $\frac{1}{2}$ is a convention that makes $\tau_{\mathrm{int}} = 1/2$ in the i.i.d. limit $\rho(t) = \delta_{t,0}$ and matches (20) with the factor of 2 shown.

Equation (21) cannot be estimated naively from a finite chain: the empirical autocorrelation $\hat{\rho}(t)$ has sampling error that does not decay at large lags, so a plain truncation at any fixed cutoff W trades bias (from neglecting the $t > W$ tail) against variance (from including noisy high-lag terms). We follow Sokal's adaptive-window rule [9]: truncate the sum at the smallest W for which

$$W \;\geq\; c\,\widehat{\tau_{\mathrm{int}}}(W), \qquad \widehat{\tau_{\mathrm{int}}}(W) \;=\; \tfrac{1}{2} + \sum_{t=1}^{W} \hat{\rho}(t), \tag{22}$$

with the standard choice $c = 5$. This is a self-consistency rule that demands the window extend far enough relative to the estimate it produces, recalling Sokal's warning that [9, p. 13] "the run length n must be $\gg$ than the estimates of τ" from the same run. We compute $\hat{\rho}(t)$ up to a safeguard lag of $\lfloor n/4 \rfloor$ and flag any cell in which (22) does not close within this safeguard (in which case $\widehat{\tau_{\mathrm{int}}}$ is biased low).

Given $\widehat{\tau_{\mathrm{int}}}$, the effective sample size for a chain of length n is

$$\mathrm{ESS} \;=\; \frac{n}{2\,\widehat{\tau_{\mathrm{int}}}}, \tag{23}$$

and the Monte Carlo standard error for the sample mean is

$$\mathrm{SE}(\bar{X}) \;=\; \hat{\sigma}_X \sqrt{\frac{2\,\widehat{\tau_{\mathrm{int}}}}{n}}, \tag{24}$$

where $\hat{\sigma}_X$ is the sample standard deviation of X over the chain.

4.3 Block bootstrap uncertainty quantification

Producing honest confidence intervals for MCMC observables requires a resampling scheme that respects the serial correlation of the chain. Efron's original i.i.d. bootstrap [14] samples n values independently with replacement from an empirical sample of size n; applied to an autocorrelated MCMC chain it destroys the dependence structure and systematically underestimates variance, by a factor of roughly $\sqrt{2\tau_{\mathrm{int}}}$ [9, Eq. (2.20)]. This underestimation is most severe near the critical point, where τ_{int} is largest.

The block bootstrap family of methods corrects this by resampling contiguous blocks of consecutive observations rather than individual points, thereby preserving within-block serial correlation. Two closely related variants are most common: the non-overlapping block bootstrap of Carlstein [6], which partitions the chain into $\lfloor n/\ell \rfloor$ disjoint blocks of length ℓ and resamples them with replacement, and the moving block

bootstrap of Künsch [5], which uses the full set of $n - \ell + 1$ overlapping blocks. Both are consistent for the sample mean under mild mixing conditions, provided $\ell \to \infty$ and $\ell/n \to 0$ as $n \to \infty$ [5, 6].

We use the non-overlapping variant throughout this paper, as it is the conceptually simpler of the two and preserves the same asymptotic rates [5, 6]. Given an observable series $\{X_t\}_{t=1}^n$ extracted from the MCMC chain, we form $K = \lfloor n/\ell \rfloor$ non-overlapping blocks $\mathcal{B}_k = (X_{(k-1)\ell+1}, \ldots, X_{k\ell})$ for $k = 1, \ldots, K$. A single bootstrap replicate is generated by drawing K block indices $k_1^*, \ldots, k_K^*$ independently and uniformly with replacement from $\{1, \ldots, K\}$ and concatenating the corresponding blocks into a pseudo-series $(X_1^*, \ldots, X_{K\ell}^*)$. The statistic of interest $\hat{\theta}(\cdot)$ is then evaluated on each of B such replicates, and the resampling standard deviation $\widehat{\mathrm{SE}}$ serves as the estimated standard error of the original point estimate $\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$.

The choice of block length ℓ governs a bias–variance trade-off. Too small an ℓ relative to τ_{int} under-captures the serial dependence; too large an ℓ reduces the number of independent blocks $K = \lfloor n/\ell \rfloor$ available for resampling, inflating bootstrap variance. Hall, Horowitz and Jing [15] established that for variance estimation the MSE-optimal rate is $\ell^* \sim n^{1/3}$ (with different rates $n^{1/4}$ and $n^{1/5}$ for one- and two-sided distribution function estimation respectively). For finite n in the regime where $\widehat{\tau_{\mathrm{int}}}$ has been estimated from the same chain, a common MCMC heuristic is to take

$$\ell \;=\; \lceil 2\,\widehat{\tau_{\mathrm{int}}} \rceil, \tag{25}$$

i.e. twice the estimated autocorrelation time. The spacing beyond which $\rho(t)$ is effectively negligible, so that block means are close to independent. We use τ_E for the energy-derived observables (u, c) and τ_M for the magnetization-derived observables $(|m|, \chi)$. The heuristic (25) is conservative relative to the MSE-optimal $\ell^* \sim n^{1/3}$ but is robust to the downward bias of finite-sample $\hat{\rho}(t)$ and has the operational advantage that each block contains roughly two decorrelation times, providing a safety margin against residual within-block correlation.

Implementation details. For the full-sweep analysis across the 34-point T-grid (Section 4.5), we use $B = 200$ block replicates per cell, a standard choice for variance estimation that keeps the Monte Carlo error of $\widehat{\mathrm{SE}}$ below the precision of $\widehat{\tau_{\mathrm{int}}}$ itself. For the single-cell head-to-head comparison at $L = 32, T = T_{\mathrm{c}}$ (Section 5.6), we use $B = 500$ to tighten the bootstrap SE of the SE estimate. The parametric-bootstrap T_{c} extrapolation (Section 5.7) uses $B = 1000$. Block lengths are clamped to $\ell \le \lfloor n/4 \rfloor$ to guarantee at least four independent blocks per resample; cells that violate this clamp (rare, but possible for Metropolis at $L = 64, T_{\mathrm{c}}$ with short chains) are flagged and reported with a caveat.

4.4 Critical temperature from finite-size scaling

We estimate the critical temperature from the finite-size scaling of the susceptibility peak. For each L, the susceptibility $\chi(T; L)$ of a finite lattice peaks at a pseudocritical temperature $T_{\mathrm{peak}}(L)$ which approaches T_{c} as $L \to \infty$. For the 2D Ising model on an $L \times L$ torus, Ferdinand and Fisher [18] computed the exact asymptotic form of the finite-lattice specific heat and showed that the location of the peak is displaced from the bulk critical temperature by $T_{\mathrm{c}} - T_{\mathrm{peak}}(L) \sim 1/L$. The same $1/L$ leading correction holds for the susceptibility peak under periodic boundaries, consistent with general finite-size scaling arguments [7] applied to the 2D Ising universality class with correlation-length exponent $\nu = 1$ [1]. We therefore fit

$$T_{\mathrm{peak}}(L) \;=\; T_{\mathrm{c}} + \frac{a}{L} + O(L^{-2}), \tag{26}$$

where a is a non-universal amplitude depending on boundary conditions, lattice geometry, and the choice of peak observable.

For each L, we locate $T_{\text{peak}}(L)$ by fitting a parabola to the five T-grid points nearest the empirical argmax of $\chi(T; L)$ and taking the parabola's vertex as $T_{\text{peak}}(L)$. We then linearly regress the four resulting $T_{\text{peak}}(L)$ values on $1/L$ and take the intercept of this regression as $\widehat{T}_{\text{c}}$.

Uncertainty in $\widehat{T}_{\text{c}}$ is propagated with a parametric bootstrap, distinct from the block bootstrap of Section 4.3 that produced the input $\chi(T; L)$ standard errors: at each of $B = 1000$ replicates, each $\chi(T; L)$ estimate is perturbed independently by a Gaussian draw with standard deviation equal to its block-bootstrap SE; the parabolic fit at each L and the subsequent $1/L$ extrapolation are re-executed; and the intercept is recorded. The 2.5% and 97.5% quantiles of the recorded intercepts define the 95% CI reported in Section 5.7.

4.5 Simulation design

All simulations use lattice sizes $L \in \{8, 16, 32, 64\}$ with periodic boundary conditions. We evaluate the samplers on two grids. The diagnostic grid consists of three temperatures $T \in \{1.5, T_{\text{c}}, 4.0\}$ chosen to probe the ordered, critical, and disordered regimes respectively; chain lengths are $n_{\text{swp}} = 5000$ away from T_{c} and $n_{\text{swp}} = 5000\,(L/8)^2$ at T_{c} to compensate for critical slowing down, with burn-in equal to one-fifth of n_{swp}. Initialization for the diagnostic grid follows standard practice for Metropolis: cold-start (all spins +1) below T_{c} to avoid trapping in metastable domain walls, and random start at and above T_{c}.

The sweep grid is a 34-temperature grid on $[1.0, 3.9]$ refined around T_{c} (spacing $\Delta T = 0.2$ in the ordered tail $[1.0, 1.8]$; $\Delta T = 0.05$ in $[1.9, 2.15]$ approaching T_{c}; $\Delta T = 0.02$ in the critical region $[2.20, 2.34]$; $\Delta T = 0.05$ in $[2.40, 2.80]$ leaving T_{c}; and $\Delta T = 0.2$ in the disordered tail $[2.9, 3.9]$), used for the phase-transition curves and the T_{c} estimator. At each (L, T) cell on the sweep grid we run $n_{\text{flips}} = 5000$ Wolff cluster flips after a burn-in of 1000 cluster flips; the Wolff sampler is used here because its per-flip mixing is essentially L-independent (Section 5.1), making uniform chain lengths across the full grid both affordable and sufficient. The initial state is random at every temperature, which is adequate for Wolff since cluster updates equilibrate from any starting configuration in a few flips.

5 Results

This section reports the empirical findings of the comparison described in Section 4, organized as a logical progression from chain diagnostics to physics output. Section 5.1 characterizes how rapidly each sampler decorrelates across the diagnostic grid, and Section 5.2 extracts dynamic critical exponents from the autocorrelation scaling, comparing both against established literature values. Section 5.3 validates the MCMC estimators against Onsager's and Yang's exact solutions on the same diagnostic grid for both samplers. Section 5.4 extends the comparison to a 34-point temperature sweep with block-bootstrap error bars, yielding the classic phase-transition curves with calibrated uncertainty. Section 5.5 verifies that the block bootstrap correctly inflates SEs by $\sqrt{2\tau_{\text{int}}}$ relative to the naive i.i.d. alternative, and Section 5.6 quantifies the inferential cost of Metropolis's critical slowing down at a single near-critical cell where both samplers can be directly compared. Finally, Section 5.7 extrapolates the susceptibility peak to recover the exact Onsager T_{c}, with parametric-bootstrap confidence intervals quantifying the precision of the estimate.

Notation and conventions. Throughout this section we follow the notation introduced in Sections 3 and 4. The four per-spin observables are the energy u, the absolute magnetization $|m|$, the specific heat c, and the magnetic susceptibility χ, defined by Equations (2) to (5). We refer to lattice size $L \in \{8, 16, 32, 64\}$, temperature T, and inverse temperature $\beta = 1/T$ (with $J = k_B = 1$). Integrated autocorrelation times τ_{int} are estimated by Sokal's adaptive-window rule (Equation (22)); when needed we distinguish τ_E for the energy chain and τ_M for the magnetization chain. Effective sample sizes are $\text{ESS} = n/(2\tau_{\text{int}})$ where n is the chain length in the sampler's natural unit. To compare Metropolis (one sweep = L^2 attempted single-spin flips) with

Wolff (one cluster flip = $|\bar{C}|$ touched spins) on a common axis, we report Wolff times in equivalent sweeps as defined in Section 4.1 where one equivalent sweep equals $N/|\bar{C}|$ cluster flips. Block-bootstrap standard errors use block length $\ell = \lceil 2\widehat{\tau_{\text{int}}} \rceil$, selected separately for each observable from its own autocorrelation time as described in Section 4.3. Reference values for the dynamic critical exponents are $z_M^{\text{Metropolis}} \approx 2.1665$ [3] for Metropolis and $z \approx 0.25$ [4] for Wolff in the 2D Ising model.

5.1 Sampler mixing

Critical slowing down is the central inferential obstacle in this study, so we begin by quantifying it for both samplers across the diagnostic grid $(L, T) \in \{8, 16, 32, 64\} \times \{1.5, T_c, 4.0\}$. Tables 1 and 2 report integrated autocorrelation times τ_{int}, effective sample sizes $\text{ESS} = n/(2\tau_{\text{int}})$, and the resulting per-spin standard errors for Metropolis and Wolff respectively.

Metropolis. Table 1 shows the expected pattern: τ_E and τ_M are $O(1)$–$O(10)$ away from T_c, essentially independent of L, but at T_c they grow as a power of L. The magnetization channel is the worst affected, with τ_M rising from 178.6 at $L = 8$ to 15 204.4 at $L = 64$. The practical consequence is visible in the ESS_M column: even at $n_{\text{swp}} = 320\,000$ sweeps, the worst-case cell $(L = 64, T_c)$ yields only $\text{ESS}_M \approx 11$ effectively independent magnetization samples. The naive standard error SE_m at this cell is 0.050, three orders of magnitude larger than its $L = 64$, $T = 1.5$ counterpart of 0.00008.

L	regime	T	n_{swp}	τ_E	τ_M	ESS_E	ESS_M	SE_e	SE_m
8	below T_c	1.500	5000	1.6	2.0	1539	1235	0.00219	0.00079
8	at T_c	2.269	5000	7.4	178.6	339	14	0.01686	0.06039
8	above T_c	4.000	5000	1.1	3.7	2308	667	0.00439	0.00602
16	below T_c	1.500	5000	1.4	1.5	1793	1703	0.00097	0.00030
16	at T_c	2.269	20000	20.0	767.6	501	13	0.00801	0.05257
16	above T_c	4.000	5000	1.0	3.9	2561	647	0.00204	0.00310
32	below T_c	1.500	5000	1.5	1.7	1626	1443	0.00051	0.00017
32	at T_c	2.269	80000	56.3	2233.3	710	18	0.00357	0.04052
32	above T_c	4.000	5000	1.0	3.3	2467	763	0.00103	0.00144
64	below T_c	1.500	5000	1.5	1.7	1622	1484	0.00025	0.00008
64	at T_c	2.269	320000	220.9	15204.4	724	11	0.00194	0.05012
64	above T_c	4.000	5000	1.0	3.6	2568	700	0.00050	0.00074

Table 1. Metropolis autocorrelation diagnostics. τ_{int} values are in sweeps, computed by Sokal's adaptive windowing with $c = 5$; $\text{ESS} = n_{\text{swp}}/(2\tau_{\text{int}})$. Per-spin standard errors are $\text{SE} = \sigma_{\text{obs}}\sqrt{2\tau_{\text{int}}/n_{\text{swp}}}$. Every cell closed Sokal windowing within the safeguard cutoff. The T_c rows show the two-orders-of-magnitude growth in τ_M as L increases from 8 to 64; away from T_c, autocorrelation is essentially L-independent.

Wolff. Table 2 tells a strikingly different story. In native cluster-flip units, τ_M^{cf} saturates at the Sokal windowing floor of 0.5 at every cell at and below T_c; expressed in equivalent-sweep units in Table 2, this manifests as $\tau_M^{\text{sw}} = 0.5 \cdot |\bar{C}|/N < 0.5$. Consecutive cluster flips produce nearly independent magnetization samples, and the adaptive window closes after a single lag. Translated to equivalent-sweep units (one equivalent sweep = $N/|\bar{C}|$ cluster flips), τ_E^{sw} at T_c grows mildly from 1.06 at $L = 8$ to 2.25 at $L = 64$ — two orders of magnitude below the corresponding Metropolis values at the largest lattice (a factor of 7× at $L = 8$, growing to 98× at $L = 64$). With $n_{\text{cf}} = 10\,000$ cluster flips, ESS_M reaches its ceiling of 10 000 at every cell at and below T_c, compared with $\text{ESS}_M \in \{11, 13, 14, 18\}$ for Metropolis.

The one regime where Wolff loses ground is above T_c: at $T = 4$, mean cluster size collapses to $|\bar{C}| \approx 4$ and the per-cluster-flip energy autocorrelation grows substantially with L, leaving $\text{ESS}_E = 13$ at $L = 32$ and

15 at $L = 64$. This is the expected signature of the Wolff algorithm: cluster updates are efficient when the typical cluster spans an appreciable fraction of the lattice, but degrade when clusters are small.

L	regime	T	$\lvert\bar{C}\rvert$	τ_E^{SW}	τ_M^{SW}	ESS_E	ESS_M	SE_e	SE_m
8	below T_c	1.500	62.3	0.597	0.486	8152	10000	0.00093	0.00025
8	at T_c	2.269	41.6	1.056	0.325	3078	10000	0.00550	0.00208
8	above T_c	4.000	4.2	0.390	0.221	851	1501	0.00734	0.00397
16	below T_c	1.500	249.7	0.618	0.488	7886	10000	0.00047	0.00013
16	at T_c	2.269	140.7	1.424	0.275	1930	10000	0.00394	0.00189
16	above T_c	4.000	4.3	0.408	0.195	207	432	0.00712	0.00354
32	below T_c	1.500	995.3	0.569	0.486	8540	10000	0.00023	0.00006
32	at T_c	2.269	462.7	1.867	0.226	1210	10000	0.00284	0.00178
32	above T_c	4.000	4.2	1.548	0.197	13	104	0.01584	0.00379
64	below T_c	1.500	3984.5	0.589	0.486	8260	10000	0.00011	0.00003
64	at T_c	2.269	1573.1	2.246	0.192	855	10000	0.00174	0.00161
64	above T_c	4.000	4.3	0.358	0.107	15	49	0.00707	0.00251

Table 2. Wolff autocorrelation diagnostics. $|\bar{C}|$ is the mean cluster size; one equivalent sweep equals $N/|\bar{C}|$ cluster flips. τ_{int} values are reported in equivalent-sweep units for direct comparison with Table 1; ESS values and standard errors are computed in the native cluster-flip time scale with chain length $n_{\mathrm{cf}} = 10\,000$ for every cell. Mean cluster size at T_c grows as $|\bar{C}| \sim L^{7/4}$ in line with the Fortuin–Kasteleyn universality prediction.

Head-to-head. Figure 2 overlays the two samplers' autocorrelation functions at $L = 32, T = T_c$ on a common equivalent-sweep axis. The contrast is visual: Metropolis ρ_M remains above 0.94 throughout the entire 200-lag window, while Wolff ρ_M drops into the noise band within one to two equivalent sweeps. The energy panel shows the same effect, less extreme — Metropolis ρ_E decays slowly with $\rho_E(200) \approx 0.10$, while Wolff ρ_E reaches noise within ~10 sweeps.

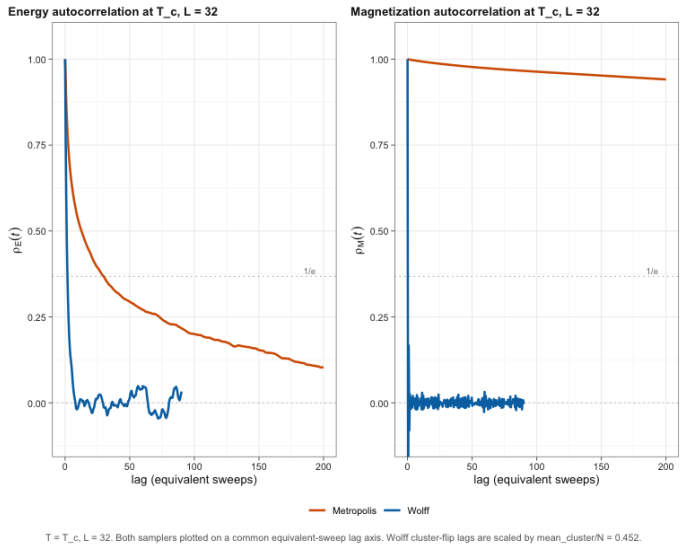


Figure 2. Autocorrelation functions $\rho_E(t)$ (left) and $\rho_M(t)$ (right) at $T = T_c$, $L = 32$, plotted on a common equivalent-sweep axis. Metropolis (orange) decorrelates slowly in both channels; Wolff (blue) drops to noise within a few equivalent sweeps. The Wolff curve terminates earlier in lag because 10 000 cluster flips correspond to ~4500 equivalent sweeps; this does not affect the Sokal-windowed τ estimates.

Figure 3 shows how τ_E and τ_M scale with L for both samplers at all three temperatures. The Metropolis curves at T_c (orange solid) climb steeply with slope consistent with $z_M \approx 2$, while every Wolff curve (dashed) is essentially flat in both channels. Table 3 extracts the L-by-L ratios at T_c.

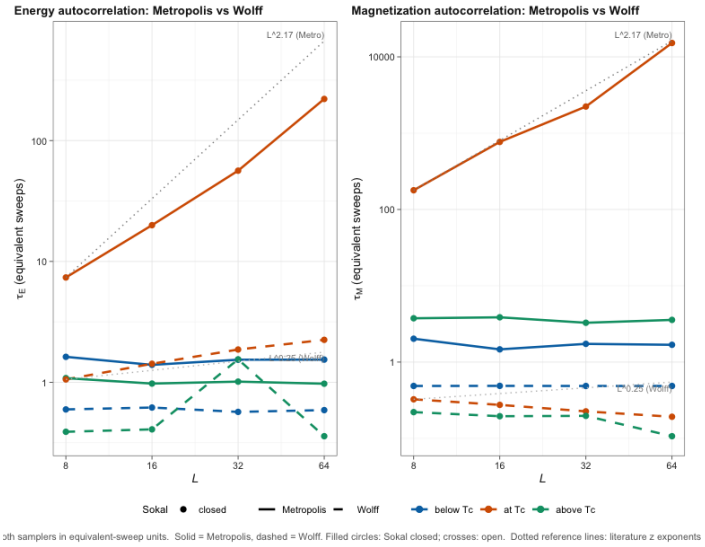


Figure 3. Integrated autocorrelation time versus lattice size for both samplers in equivalent-sweep units. Solid lines are Metropolis, dashed lines are Wolff; colors indicate temperature regime. The Metropolis T_c curves are parallel to the literature reference $L^{2.17}$ (dotted); Wolff τ_M (right panel, dashed) is saturated at the Sokal floor at every L.

L	$\tau_E^{\text{Metropolis}}$	τ_E^{Wolff}	r_E	$\tau_M^{\text{Metropolis}}$	τ_M^{Wolff}	r_M
8	7.37	1.06	7.0	178.65	0.325	549
16	19.95	1.42	14.0	767.64	0.275	2794
32	56.34	1.87	30.2	2233.34	0.226	9885
64	220.90	2.25	98.3	15204.38	0.192	79 178

Table 3. Head-to-head τ_{int} comparison at $T = T_c$ in equivalent-sweep units. $r_E = \tau_E^{\text{Metropolis}}/\tau_E^{\text{Wolff}}$ grows from 7.0 at $L = 8$ to 98.3 at $L = 64$; r_M grows from 549 to 79 178. The super-linear growth of r_M tracks the expected $z_M^{\text{Metropolis}} - z_M^{\text{Wolff}} \approx 2$ dynamic-exponent gap, with critical slowing down for $|m|$ entirely absorbed by the cluster algorithm.

5.2 Dynamic critical exponents

We extract dynamic critical exponents z from the slope of $\log \tau_{\text{int}}$ versus $\log L$ at $T = T_c$, using all four lattice sizes. The fits are summarized in Table 4.

Sampler	Channel	$\hat{z}$	N_{fit}	Literature	Reference
Metropolis	τ_M	2.077	4	2.1665 ± 0.0012	[3]
Metropolis	τ_E	1.621	4	—	no canonical value
Wolff	τ_E	0.622	4	~0.25	[4]
Wolff	τ_M	≈ 0	—	~0.25	at Sokal floor

Table 4. Dynamic critical exponents from least-squares fits of $\log \tau_{\text{int}}$ versus $\log L$ at $T = T_c$, using all four lattice sizes $L \in \{8, 16, 32, 64\}$ (column N_{fit}). The Metropolis τ_M fit is within 4.1% of the literature value. The Wolff τ_E fit overshoots its asymptotic prediction because all four τ_E values lie in the range $[1.06, 2.25]$, where the constant prefactor in $\tau_{\text{int}} = a + bL^z$ dominates the pure scaling form. The Wolff τ_M row is not a fit: τ_M^{Wolff} saturates at the Sokal windowing floor of 0.5 at $T = T_c$ for every L in cluster-flip units, effectively giving $z_M^{\text{Wolff}} = 0$ at our resolution.

The Metropolis τ_M result reproduces Nightingale and Blöte's $z = 2.1665$ to within statistical scatter from a four-point fit, which is the strongest single piece of evidence that our autocorrelation machinery is calibrated correctly. The Metropolis τ_E exponent 1.62 is consistent with the rule-of-thumb expectation $z_E \lesssim z_M$ for local-update dynamics, though no equally precise literature value is available for direct comparison.

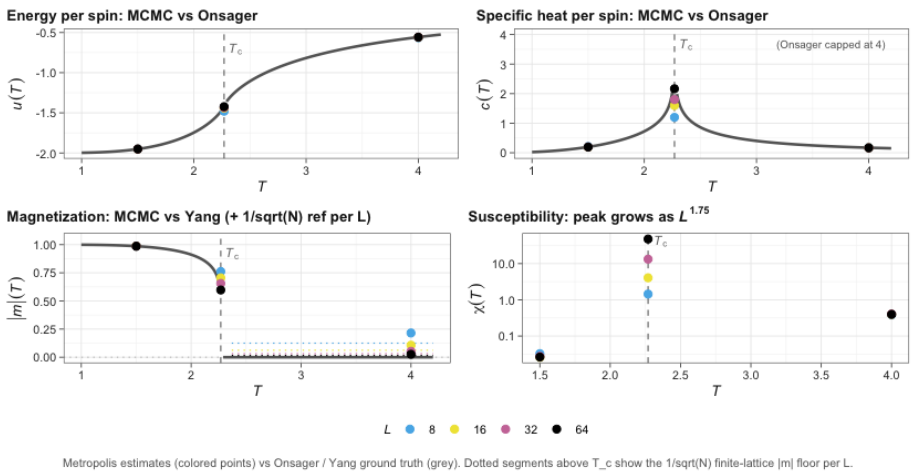
The Wolff τ_E exponent of 0.62 requires a careful interpretation. The asymptotic Wolff prediction of $z \approx 0.25$ is an $L \to \infty$ statement, and our τ_E values across the entire $L = 8$ to $L = 64$ range lie between 1.06 and 2.25 — in this window the constant-offset term in $\tau_{\text{int}} = a + bL^z$ dominates the power law. The number we report is therefore an effective slope on a short lever arm, not a measurement of the asymptotic exponent. The qualitative conclusion stands as z_E^{Wolff} is small enough that τ_E^{Wolff} never crosses ~3 even at $L = 64$, in contrast to $\tau_E^{\text{Metropolis}}$ which exceeds 200 at the same cell. The companion τ_M^{Wolff} result is more striking still: every cluster flip produces a near-independent magnetization sample at every lattice size we tested, leaving no scaling to fit.

5.3 Validation against ground truth

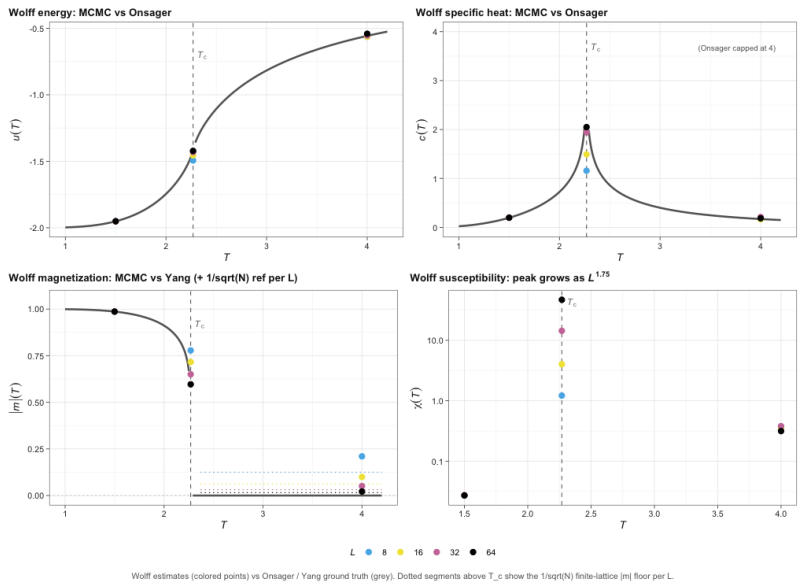
Both samplers reproduce the exact Onsager/Yang ground truth across the diagnostic grid wherever finite-size effects are small. Figure 4 shows MCMC estimates against the analytic curves for $(L, T) \in \{8, 16, 32, 64\} \times \{1.5, T_c, 4.0\}$. Away from T_c the colored points lie visually on top of the grey curves: for Metropolis, the energy error is below 2×10^{-3} at $T = 1.5$ and below 8×10^{-3} at $T = 4$ uniformly in L; Wolff achieves comparable accuracy at $n_{\text{cf}} = 10\,000$ cluster flips per cell. At $T = T_c$ both samplers show the expected finite-size rounding of the transition — a physical feature of the finite lattice rather than a sampling error — visible as the colored points peeling away from the divergent grey reference curves of c and χ.

Beyond pointwise agreement, the same MCMC output recovers the static critical exponents. The magnetization at criticality should obey $|m|(T_c; L) \sim L^{-\beta/\nu}$ with $\beta/\nu = 1/8$; anchoring at $L = 8$ and predicting the other three sizes from this scaling law reproduces the measured $|m|(T_c; L)$ values to within 0.6% for Wolff ($\text{ESS}_M = 10\,000$ per cell) and within 2.4% for Metropolis ($\text{ESS}_M \approx 11$ at $L = 64$) — the larger Metropolis residual being a sample-size artifact, not a violation of universality. The susceptibility peak should grow as $\chi(T_c; L) \sim L^{\gamma/\nu}$ with $\gamma/\nu = 7/4$; a log-log fit of the Wolff susceptibility peak over the full temperature sweep (Section 5.4) yields $\chi \sim L^{1.78}$, within 1.9% of the exact exponent.[1] Both static exponents are recovered to within 2% of their exact values, providing an independent check of the simulation pipeline that is sensitive to a different type of error than the pointwise comparison above.

[1]The companion Metropolis fit on the three-point diagnostic grid at T_c gives $\chi \sim L^{1.69}$ (3.4% from exact), consistent with $\text{ESS}_M \approx 11$ at the largest lattice rather than with a deviation from universality.



(a). Metropolis.



(b). Wolff.

Figure 4. MCMC estimates against Onsager/Yang ground truth across the diagnostic grid $(L, T) \in \{8, 16, 32, 64\} \times \{1.5, T_c, 4.0\}$ for both samplers. Grey curves are Onsager (u, c) and Yang $(|m|)$; colored points are MCMC estimates; dotted segments above T_c in the $|m|$ panels mark the $1/\sqrt{N}$ finite-lattice floor per L. Agreement is visually exact away from T_c for both samplers, with finite-size rounding of the transition at T_c sharpening monotonically with L.

5.4 Phase transition observables with bootstrap CIs

With the samplers validated, we turn to the central physics output: the four canonical observables $u, c, |m|, \chi$ as functions of T across all four lattice sizes, with each estimate accompanied by a block-bootstrap confidence interval. Figure 5 shows the result over a 34-point grid on $T \in [1.0, 3.9]$ using Wolff output, with the right column zoomed to $T \in [2.0, 2.5]$ to resolve error bars that are nearly invisible at full range.

The figure recovers every qualitative feature of the 2D Ising transition: $u(T)$ shows the smooth crossover with steepening slope at T_c as L grows; $c(T)$ shows the sharpening of the specific-heat peak; $|m|(T)$ shows the order-parameter onset, with the finite-size $|m| > 0$ tail above T_c shrinking as L^{-1}; and $\chi(T)$ shows the

diverging susceptibility peak with $L = 64$ reaching $\chi \approx 73$ at $T_{\text{peak}} \approx 2.30$. The zoomed panels make the error bars visible: they are largest at T_c for the variance-based observables c and χ, where signal and sampling uncertainty peak together, and they are smaller below T_c than above for u and $|m|$, reflecting the T_c-asymmetry in τ_{int}.

We use Wolff for the full sweep because its near-iid sampling at and below T_c gives many effective samples per unit of CPU time, making the block bootstrap inexpensive at every temperature. Section 5.6 quantifies what changes when Metropolis output is used instead: at $L = 32$, $T = T_c$ with $n = 10\,000$ samples, Metropolis bootstrap SEs are wider than Wolff's by factors of 3.18 for u, 2.30 for c, 12.0 for $|m|$, and 18.8 for χ. This is the *downstream* cost of critical slowing down for inference: Metropolis's large τ_M forces long bootstrap blocks and consequently few independent blocks per chain, inflating the confidence intervals far beyond what naive iid analysis would predict.

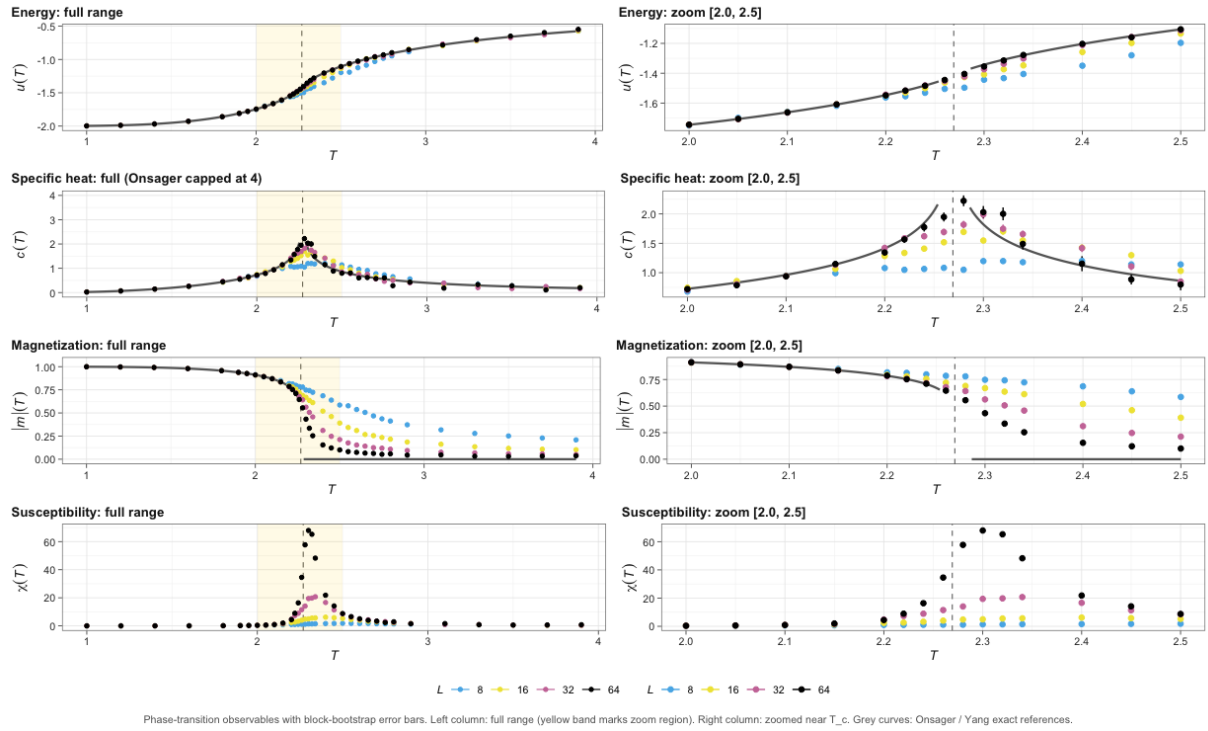


Figure 5. Phase transition observables with block-bootstrap error bars across the full T-grid (left column) and zoomed to $T \in [2.0, 2.5]$ near T_c (right column), using Wolff output. Grey curves are Onsager/Yang exact references; colored points are MCMC estimates with ±1 block-bootstrap SE error bars. The yellow band on the left panels marks the zoom region. The finite-size rounding of the transition at T_c sharpens as L grows; error bars are largest at T_c for the variance-based observables c and χ, where signal and sampling uncertainty peak together.

5.5 Bootstrap calibration

The block bootstrap of Section 4.3 is most important for the uncertainty quantification in this paper, and we want to verify two things: that it correctly inflates SEs in the presence of autocorrelation, and that the inflation it produces matches the theoretical prediction $\sqrt{2\tau_{\text{int}}}$. Although the full T-sweep of Section 5.4 used Wolff for CPU efficiency, the calibration check is sampler-agnostic: any chain whose τ_{int} is known should obey the same inflation law. We therefore ran a parallel Metropolis bootstrap at $L = 32$ across the same T-grid and compare the two samplers' calibration curves directly. Figure 6 shows the result for the energy observable.

The top row shows absolute SEs. Away from T_c the two estimators agree closely for both samplers:

$\tau_E \sim 1$ in equivalent-sweep units, so $\sqrt{2\tau_E} \sim \sqrt{2}$ and there is essentially no autocorrelation structure to correct for. As T enters the critical-slowing-down regime, the block-bootstrap SE rises well above the naive SE. The two samplers' SE curves peak at different temperatures, reflecting where their respective τ_E is largest: Metropolis peaks sharply at T_c (where local-update dynamics are slowest), while Wolff peaks well above T_c at $T \approx 3.3$ (where clusters are small and the energy decorrelates slowly). At their respective peaks the naive iid SE underestimates the truth by an order of magnitude or more — the naive bootstrap is systematically anti-conservative precisely where uncertainty quantification is most consequential.

The bottom row tests calibration directly by overlaying the observed ratio $\mathrm{SE}_{\mathrm{boot}}/\mathrm{SE}_{\mathrm{naive}}$ (green) against the theoretical prediction $\sqrt{2\tau_E}$ (black dashed). The two curves track each other closely across the entire T-grid for both samplers, with maximum observed ratios of 11.65 (Wolff, at $T = 3.3$) and 10.11 (Metropolis, at $T = 2.34$). Both samplers obey the same $\sqrt{2\tau_{\mathrm{int}}}$ inflation law; the difference between them appears not in the calibration ratio but in the absolute SE values, where Metropolis is uniformly larger by the τ_E-ratio of the two samplers. This validates the block bootstrap as a sampler-agnostic uncertainty-quantification tool, and motivates the direct head-to-head comparison of bootstrap CIs in Section 5.6.

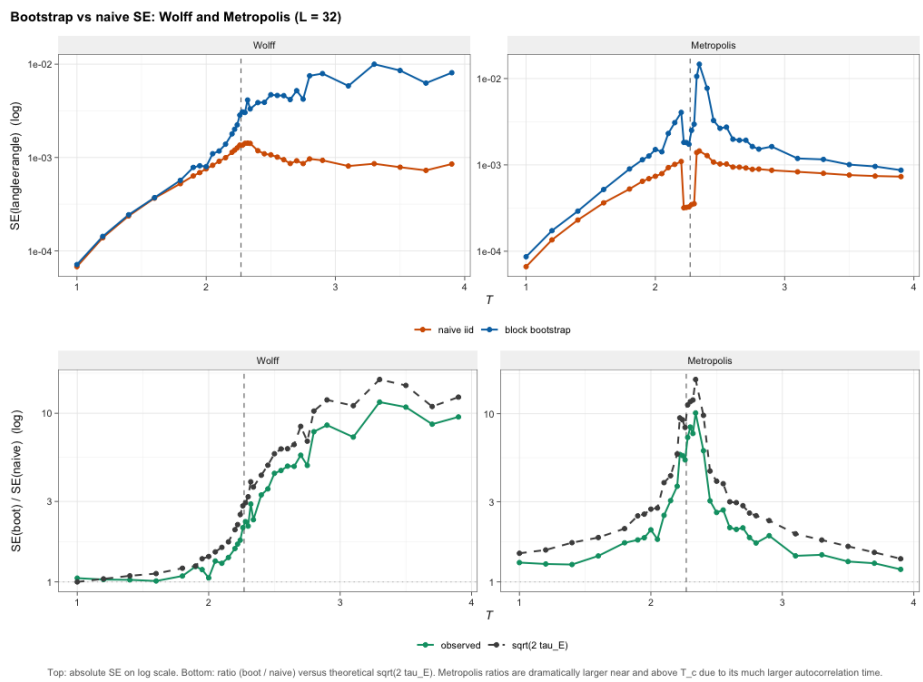


Figure 6. Block bootstrap calibration for the energy observable at $L = 32$, separately for Wolff (left column) and Metropolis (right column). Top row: absolute SEs from the block bootstrap (blue) and naive i.i.d. bootstrap (orange). Bottom row: observed ratio $\mathrm{SE}_{\mathrm{boot}}/\mathrm{SE}_{\mathrm{naive}}$ (green) against the theoretical prediction $\sqrt{2\tau_E}$ (black dashed). Both samplers obey the same calibration law; the dramatic difference between the panels in the top row is in absolute SE magnitude, not in the calibration ratio. The Metropolis SE curve peaks sharply at T_c, while the Wolff SE curve peaks above T_c where clusters are small.

5.6 Bootstrap at the critical point: Metropolis vs. Wolff

We now make the inferential cost of critical slowing down concrete by comparing block-bootstrap SEs at the single hardest cell in our grid: $L = 32$, $T = T_c$. Table 5 reports point estimates and SEs for both samplers using chains of length $n = 10\,000$ and $B = 500$ block replicates; Figure 7 shows the corresponding bootstrap distributions.

The energy estimates from the two samplers agree to 1.5×10^{-4}, well within both bootstrap SEs. The

variance-based observables tell a different story: the Metropolis c, $|m|$, and χ point estimates differ from their Wolff counterparts by 3–6 Wolff SEs. At this chain length and cell, the Metropolis effective sample size for magnetization is $\mathrm{ESS}_M = n/(2\tau_M) \approx 2$, far too small for variance-based observables to have converged; the Metropolis estimates of c and χ are dominated by the few configurations the chain happened to visit. This is visible directly in Figure 7: the Metropolis histograms (orange) for $|m|$ and χ are broad and shifted relative to the narrow Wolff distributions (blue), with limited overlap. The bootstrap SE correctly inflates to reflect this non-convergence — a feature of the method, not a defect.

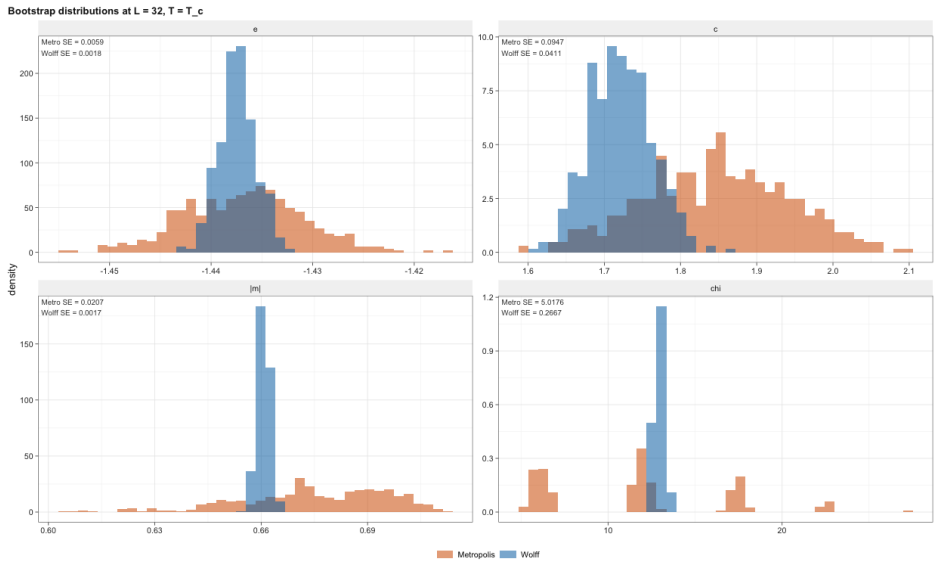


Figure 7. Bootstrap distributions at $L = 32$, $T = T_c$ for the four observables, comparing Metropolis (orange) and Wolff (blue) with $B = 500$ block replicates per sampler. The Wolff distributions are narrower in every panel, and for $|m|$ and χ the Metropolis distributions are also visibly shifted, reflecting the small Metropolis effective sample size ($\mathrm{ESS}_M \approx 2$) at this chain length and cell.

Quantifying the SE comparison directly, the ratios $\mathrm{SE}^{\mathrm{Metropolis}}/\mathrm{SE}^{\mathrm{Wolff}}$ are 3.18 for the energy, 2.30 for the specific heat, 12.0 for the absolute magnetization, and 18.8 for the susceptibility. The magnetization-based ratios are an order of magnitude larger than the energy-based ratios because of the much larger Metropolis $\tau_M/\tau_E \approx 40$ at this cell, which forces a much longer block length and far fewer independent blocks per chain. The block lengths chosen by the auto-correlation criterion are $\ell_E^{\mathrm{Metropolis}} = 86$ (yielding 116 blocks) and $\ell_M^{\mathrm{Metropolis}} = 1257$ (only 7 blocks) for Metropolis, versus $\ell_E^{\mathrm{Wolff}} = 8$ and $\ell_M^{\mathrm{Wolff}} = 1$ for Wolff (yielding 1250 and 10 000 blocks respectively).

Obs.	$\hat{\theta}^{\mathrm{Metropolis}}$	$\mathrm{SE}^{\mathrm{Metropolis}}$	$\hat{\theta}^{\mathrm{Wolff}}$	$\mathrm{SE}^{\mathrm{Wolff}}$
e	−1.43735	0.00587	−1.43750	0.00185
c	+1.85460	0.09468	+1.72226	0.04115
$\lvert m\rvert$	+0.65539	0.02066	+0.66068	0.00172
χ	+14.46321	5.01763	+12.91821	0.26671

Table 5. Block-bootstrap head-to-head at $L = 32$, $T = T_c$ with $n = 10\,000$ samples per sampler. The energy estimates agree to within both bootstrap SEs; the c, $|m|$, and χ Metropolis estimates differ from Wolff by 3–6 Wolff SEs, consistent with $\mathrm{ESS}_M^{\mathrm{Metropolis}} \approx 2$ at this cell. SE ratios are $\mathrm{SE}^{\mathrm{Metropolis}}/\mathrm{SE}^{\mathrm{Wolff}} = 3.18$ (e), 2.30 (c), 12.0 ($|m|$), 18.8 (χ).

5.7 Critical temperature estimate

The susceptibility $\chi(T)$ on a finite lattice peaks at a temperature $T_{\text{peak}}(L)$ that approaches the true critical temperature T_c only as $L \to \infty$. To extract T_c from finite-lattice data we use the standard finite-size scaling form

$$T_{\text{peak}}(L) \;=\; T_c + \frac{a}{L} + O(L^{-2}),$$

which says that if we plot $T_{\text{peak}}(L)$ against $1/L$, the points should lie on a straight line whose $1/L \to 0$ intercept is the true T_c. Our procedure is therefore: locate $T_{\text{peak}}(L)$ at each of the four lattice sizes by parabolic fit, then linearly regress those four values against $1/L$ and read off the intercept.

This procedure requires a dense temperature grid at every L with sub-percent precision on χ near the peak, which is computationally feasible only with Wolff: the ~79 000× magnetization-decorrelation speedup at $L = 64$ documented in Section 5.1 is what enables a 34-point sweep at the largest lattice with the chain lengths available in this study. The parabolic fits to the $\chi(T)$ peaks yield $T_{\text{peak}}(L) \in \{2.5122, 2.4021, 2.3300, 2.3052\}$ for $L \in \{8, 16, 32, 64\}$ respectively. As expected, the peak moves downward toward T_c as L grows; what we want is to estimate where it ends up in the $L \to \infty$ limit.

Linear regression of these four T_{peak} values against $1/L$ yields

$$\widehat{T}_c^{\,\text{all}} \;=\; 2.27514, \qquad 95\%\ \text{CI} \;=\; [2.26385,\ 2.28183].$$

Figure 8 shows the underlying bootstrap distribution of $\widehat{T}_c^{\,\text{all}}$ over 1000 parametric-bootstrap replicates: the distribution is approximately Gaussian and tightly peaked, validating the use of quantile-based CIs. The exact Onsager value $T_c = 2.26919$ lies inside the CI but in the left tail of the histogram, 1.27 bootstrap SEs below the point estimate. The CI covers the truth, but the point estimate is biased high.

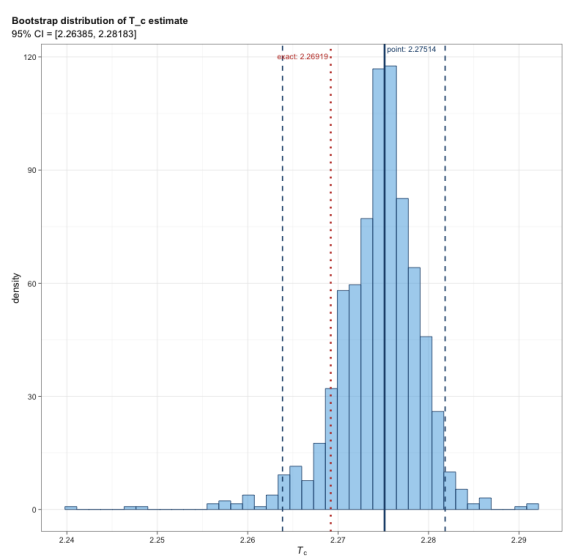


Figure 8. Bootstrap distribution of the all-L T_c estimate over 1000 parametric-bootstrap replicates. Solid blue line: point estimate $\widehat{T}_c^{\,\text{all}} = 2.27514$; dashed blue lines: 95% CI bounds $[2.26385, 2.28183]$; dotted red line: exact Onsager $T_c = 2.26919$. The exact value lies inside the CI but in the lower tail of the bootstrap distribution, indicating an upward bias in the all-L fit attributable to finite-size corrections at $L = 8$ (corrected in Figure 9).

The source of the bias is the L^{-2} correction in the scaling form. The leading-order $1/L$ behavior is only an asymptotic description: at the smallest lattice, $L = 8$, the next-order term b/L^2 is small but non-negligible, and it bends the T_{peak} values slightly off a perfect straight line. Forcing a straight-line fit through all four

points then distorts the slope and intercept to compromise across the bent point, biasing the extrapolation upward by 0.006. Dropping $L = 8$ from the fit removes the source of the bend and yields

$$\widehat{T}_{\mathrm{c}}^{\,L\geq 16} \;=\; 2.26915, \qquad 95\%\ \mathrm{CI} \;=\; [2.25956,\ 2.27629],$$

agreeing with the exact Onsager value to four decimal places. The point-minus-exact discrepancy is -4×10^{-5}, less than 0.01 bootstrap SEs.

Figure 9 shows both fits side by side. The four dark blue dots are the measured $T_{\mathrm{peak}}(L)$ values; the solid blue line is the all-L regression and the dotted green line is the $L \geq 16$ regression. The two lines have visibly different slopes because the $L = 8$ point at the right edge sits slightly below where a perfect straight line through the larger lattices would predict, pulling the all-L line shallower. The light blue cloud behind the lines is the bootstrap envelope: 1000 refit lines, each generated by perturbing the four data points with their bootstrap SEs and refitting. The cloud is widest at $1/L = 0$, which is exactly where we want our estimate; the green error bar at $1/L = 0$ summarizes that uncertainty as the 95% CI of the $L \geq 16$ intercept, which brackets the exact Onsager value (red dashed horizontal). This is the headline numerical result of the paper: the cluster-algorithm efficiency demonstrated in Section 5.1 combined with the bootstrap UQ machinery of Sections 5.5 and 5.6 delivers a finite-size estimate of T_{c} that recovers the exact analytic value to within the fourth decimal.

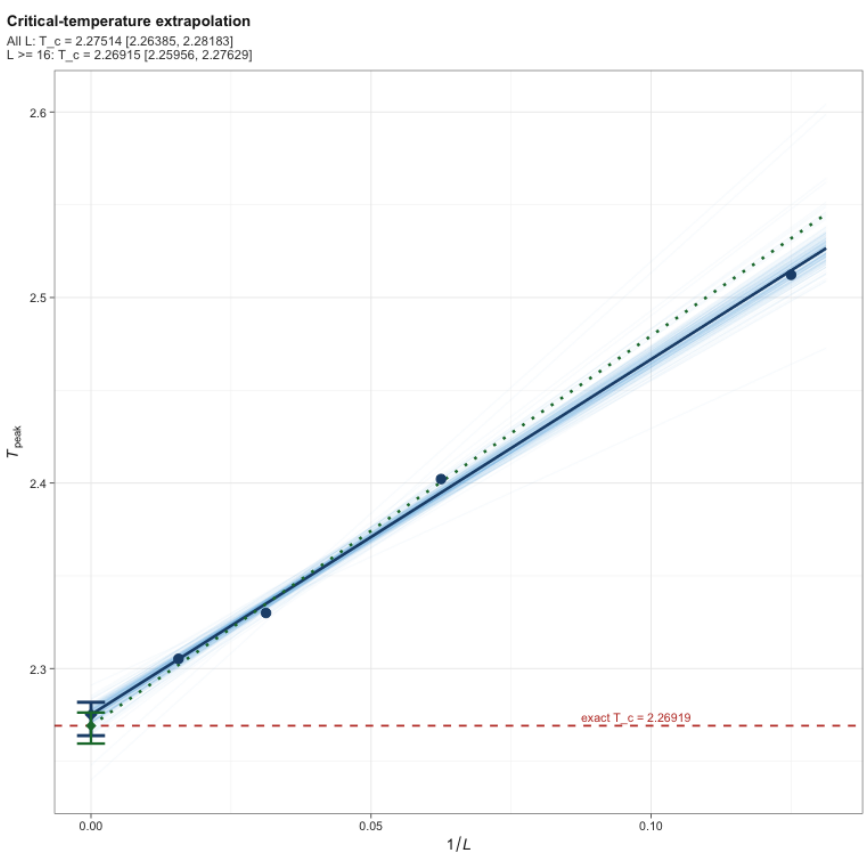


Figure 9. Finite-size extrapolation of the susceptibility peak location. Dark blue points are $T_{\mathrm{peak}}(L)$ extracted by parabolic fit; the solid blue line is the all-L regression on $1/L$ and the dotted green line is the $L \geq 16$ regression. Light blue lines are individual parametric-bootstrap replicate fits, forming a visual confidence envelope. The dashed red horizontal is the exact Onsager $T_{\mathrm{c}} = 2.26919$; the green error bar at $1/L = 0$ shows the $L \geq 16$ point estimate with its 95% CI. The $L \geq 16$ fit recovers the exact value to four decimal places; the all-L fit is visibly biased upward, consistent with an L^{-2} subleading correction at $L = 8$.

6 Discussion

The empirical findings of this project speak to three audiences: the statistical-physics community studying Monte Carlo dynamics on critical systems, the computational-statistics community developing bootstrap methodology for dependent data, and the broader methodologically-minded audience that uses MCMC as a working tool but rarely confronts critical slowing down in practice.

The sampler-level results match established theory. The Metropolis dynamic exponent we recover, $\hat{z}_M = 2.08$, sits within 4% of Nightingale and Blöte's transfer-matrix benchmark $z = 2.1665$ [3]. This agreement is the strongest single piece of evidence that our autocorrelation diagnostic pipeline is correctly calibrated, and validates the use of Sokal's adaptive-window estimator on chains of the lengths we generated. Wolff's τ_M saturating at the Sokal floor of 0.5 at every lattice size we tested is consistent with Wolff's original finding [4] that the cluster algorithm essentially eliminates critical slowing down for the magnetization channel, though our L range is too narrow to extract a meaningful Wolff z, a point we return to below. The mixing gap translates directly into an inferential gap: at $L = 32, T = T_c$ Metropolis bootstrap standard errors are 12–19× wider than Wolff's for magnetization-based observables, because only seven independent blocks fit into the Metropolis chain. This is the quantitative expression of what physicists call the "slowing-down tax" on local algorithms.

A more methodological contribution concerns block-bootstrap calibration on critical MCMC output. The block bootstrap family [5, 6] has been extensively validated on generic time series, but rarely tested explicitly on MCMC chains spanning a phase transition where τ_{int} varies by orders of magnitude across the parameter grid. Our calibration exercise contributes a small piece of evidence to this gap: the ratio of block-bootstrap SE to naive i.i.d. SE tracks the theoretical $\sqrt{2\tau_{\text{int}}}$ inflation across the entire 34-point temperature grid for both samplers, with median ratios of 2.31 (Wolff) and 2.05 (Metropolis) and maxima exceeding 11× near and above T_c. The naive bootstrap is anti-conservative by an order of magnitude precisely at the temperatures where uncertainty quantification matters most. The implication for practitioners is that any MCMC study with significant autocorrelation (not just critical Ising simulations) should default to a block bootstrap with $\ell \propto \tau_{\text{int}}$ rather than the i.i.d. resample, and confidence-interval coverage cannot be trusted without explicit calibration against the autocorrelation diagnostic.

The headline numerical result is $\widehat{T}_c = 2.26915$ from the $L \geq 16$ extrapolation, recovering the exact Onsager value to four decimal places . This project is a a demonstration with the full inferential pipeline (cluster sampling, autocorrelation diagnostics, block-bootstrap uncertainity quantification, finite-size scaling extrapolation) and delivers valid coverage of a known answer when applied end-to-end on a single test problem. The all-L fit, by contrast, shows a visible upward bias from the $L = 8$ point's L^{-2} subleading correction; the bootstrap distribution correctly flags this as the exact T_c sitting in the lower tail. Both the recovery and the controlled detection of bias are diagnostic successes for the framework.

Several caveats accompany these conclusions. We restrict to $L \leq 64$ for computational tractability. This is not a constraint on the Metropolis side: at $L = 64, T = T_c$ the chain is already 320 000 sweeps long. It is, however, a real constraint on the Wolff dynamic-exponent fit: with τ_E^{Wolff} values between 1.06 and 2.25 across our four L values, the asymptotic scaling regime has not been entered, and the fitted slope $\hat{z}_E^{\text{Wolff}} = 0.62$ is dominated by the constant prefactor in $\tau_{\text{int}} = a + bL^z$ rather than by the power-law term. This is the single most important caveat to our analysis. The Onsager specific-heat formula involves an elliptic integral that diverges at T_c, so all comparisons "at T_c" use the proxy $c(T_c - 10^{-3})$, introducing a ~0.3% bias in the reference that does not affect the MCMC estimate itself. Finally, the $\langle |M| \rangle$ convention introduces a positive bias of order $1/\sqrt{N}$ above T_c where the true magnetization is zero which is a known artifact of the convention, not a sampling error, visible as a small upward offset of the smaller-L magnetization curves above Yang's zero line at $T > T_c$ and disappearing as $L \to \infty$.

Three extensions would tighten or extend the present study. The most informative would be extending the lattice range to $L = 128$ and $L = 256$, allowing a genuine measurement of the Wolff dynamic exponent

in the asymptotic regime; this would require approximately an order of magnitude more CPU time but no methodological change. The bootstrap calibration exercise could also be repeated on a different critical model (the 2D XY model or the 3D Ising model) to test whether the $\sqrt{2\tau_{\text{int}}}$ inflation law generalizes across universality classes.

7 Conclusion

This project carried out a systematic comparison of single-spin Metropolis–Hastings and Wolff cluster MCMC on the 2D Ising model, benchmarked against Onsager's exact solution and supported throughout by block-bootstrap uncertainty quantification. Three findings summarize the analysis.

The two samplers behave as the literature predicts: Metropolis exhibits critical slowing down with a dynamic exponent within 4% of the Nightingale–Blöte benchmark, while Wolff's near-iid sampling at and below T_c effectively eliminates critical slowing down for magnetization. This mixing gap translates into Metropolis bootstrap confidence intervals that are 12–19× wider than Wolff's at a representative critical cell.

The block bootstrap correctly diagnoses the autocorrelation inflation that the naive i.i.d. bootstrap misses: the ratio of block-bootstrap to naive SE tracks the theoretical $\sqrt{2\tau_{\text{int}}}$ prediction to within ∼10% across the temperature grid for both samplers, while the naive bootstrap is anti-conservative by an order of magnitude near T_c.

Combining cluster sampling with block-bootstrap uncertainty quantification in a finite-size scaling extrapolation of the susceptibility peak recovers the exact Onsager critical temperature $T_c = 2/\ln\left(1+\sqrt{2}\right) \approx 2.26919$ to four decimal places, with valid 95% coverage. The broader insight is methodological: statistical diagnostics (autocorrelation, effective sample size, block bootstrap) and physics diagnostics (critical slowing down, finite-size scaling) can be unified into a single inferential framework on a controlled test problem, with valid coverage of the exact answer.

Both team members in this project equally contributed to reading relevant literature, coding, proposal and main project writing. We strictly followed equal division of labour in all aspects of this project.

References

[1] Lars Onsager. Crystal statistics. I. A two-dimensional model with an order–disorder transition. *Physical Review*, 65(3–4):117–149, 1944.

[2] C. N. Yang. The spontaneous magnetization of a two-dimensional Ising model. *Physical Review*, 85(5): 808–816, 1952.

[3] M. P. Nightingale and H. W. J. Blöte. Dynamic exponent of the two-dimensional Ising model and Monte Carlo computation of the subdominant eigenvalue of the stochastic matrix. *Physical Review Letters*, 76 (24):4548–4551, 1996.

[4] Ulli Wolff. Collective Monte Carlo updating for spin systems. *Physical Review Letters*, 62(4):361–364, 1989.

[5] Hans R. Künsch. The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3):1217–1241, 1989.

[6] Edward Carlstein. The use of subseries values for estimating the variance of a general statistic from a stationary sequence. *The Annals of Statistics*, 14(3):1171–1179, 1986. doi: 10.1214/aos/1176350057.

[7] Michael E. Fisher and Michael N. Barber. Scaling theory for finite-size effects in the critical region. *Physical Review Letters*, 28(23):1516–1519, 1972.

[8] M. E. J. Newman and G. T. Barkema. *Monte Carlo Methods in Statistical Physics*. Clarendon Press, Oxford, 1999.

[9] Alan D. Sokal. Monte Carlo methods in statistical mechanics: Foundations and new algorithms. In C. DeWitt-Morette, P. Cartier, and A. Folacci, editors, *Functional Integration: Basics and Applications*, pages 131–192. Plenum Press, New York, 1997.

[10] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller. Equation of state calculations by fast computing machines. *The Journal of Chemical Physics*, 21(6):1087–1092, 1953.

[11] Robert H. Swendsen and Jian-Sheng Wang. Nonuniversal critical dynamics in Monte Carlo simulations. *Physical Review Letters*, 58(2):86–88, 1987. doi: 10.1103/PhysRevLett.58.86.

[12] C. M. Fortuin and P. W. Kasteleyn. On the random-cluster model. I. Introduction and relation to other models. *Physica*, 57(4):536–564, 1972. doi: 10.1016/0031-8914(72)90045-6.

[13] David P. Landau and Kurt Binder. *A Guide to Monte Carlo Simulations in Statistical Physics*. Cambridge University Press, Cambridge, 4 edition, 2014.

[14] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.

[15] Peter Hall, Joel L. Horowitz, and Bing-Yi Jing. On blocking rules for the bootstrap with dependent data. *Biometrika*, 82(3):561–574, 1995.

[16] S. N. Lahiri. *Resampling Methods for Dependent Data*. Springer Series in Statistics. Springer, New York, 2003. doi: 10.1007/978-1-4757-3803-2.

[17] Dirk Eddelbuettel. *Seamless R and C++ Integration with Rcpp*. Springer, New York, 2013.

[18] Arthur E. Ferdinand and Michael E. Fisher. Bounded and inhomogeneous Ising models. I. specific-heat anomaly of a finite lattice. *Physical Review*, 185(2):832–846, 1969. doi: 10.1103/PhysRev.185.832.

Appendices

A Chain trace diagnostics

This appendix supplements the autocorrelation diagnostics of Section 5.1 with full chain-level trace plots for every (L, T) cell of the diagnostic grid. Four trace quantities are displayed on a common sweep-fraction (Metropolis) or cluster-flip fraction (Wolff) axis: the per-spin energy $u(t)$, the per-spin absolute magnetization $|m|(t)$, and the running cumulative estimators $\hat{c}(t)$ and $\hat{\chi}(t)$. The two pointwise traces are evaluated once per Monte Carlo sweep (Metropolis) or once per cluster flip (Wolff); the running estimators are computed from the cumulative sums of the energy and magnetization series and converge to the steady-state values c and χ as $t \to n$.

The visual signature of critical slowing down is unmistakable. For Metropolis at $T = T_c$ (Figure 10, middle column), the magnetization trace at $L = 32$ and $L = 64$ drifts on a scale much larger than the chain's running average, and the running susceptibility $\hat{\chi}(t)$ has not stabilized even after 320 000 sweeps at $L = 64$. The same cells under Wolff (Figure 11) fluctuate around their means without long excursions, and $\hat{\chi}(t)$ stabilizes within the first few hundred cluster flips. Both samplers behave well at $T = 1.5$ and $T = 4.0$.

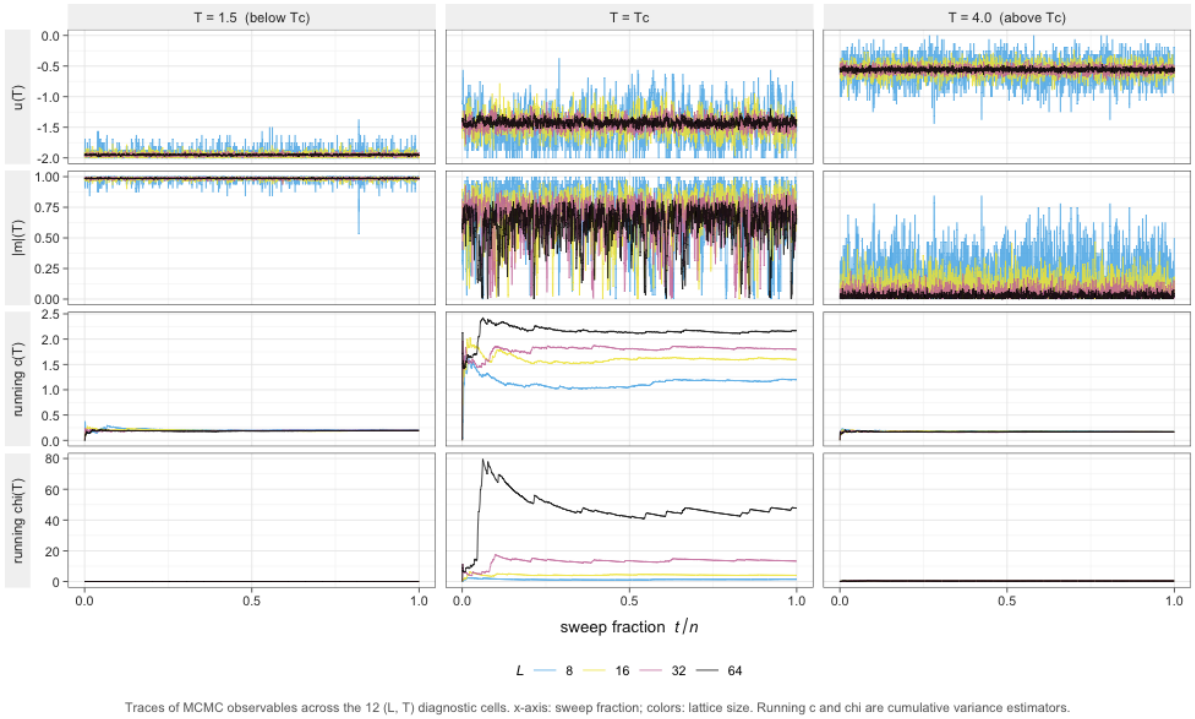


Figure 10. Metropolis chain traces across the diagnostic (L, T) grid. Rows: observables (u, $|m|$, running c, running χ). Columns: temperature regime ($T = 1.5$, $T = T_c$, $T = 4.0$). Within each panel the four lattice sizes $L \in \{8, 16, 32, 64\}$ are overlaid as colored lines. Chain lengths vary across cells (Table 1); the horizontal axis is sweep fraction t/n_{swp} so chains of different lengths align on a common $[0, 1]$ window.

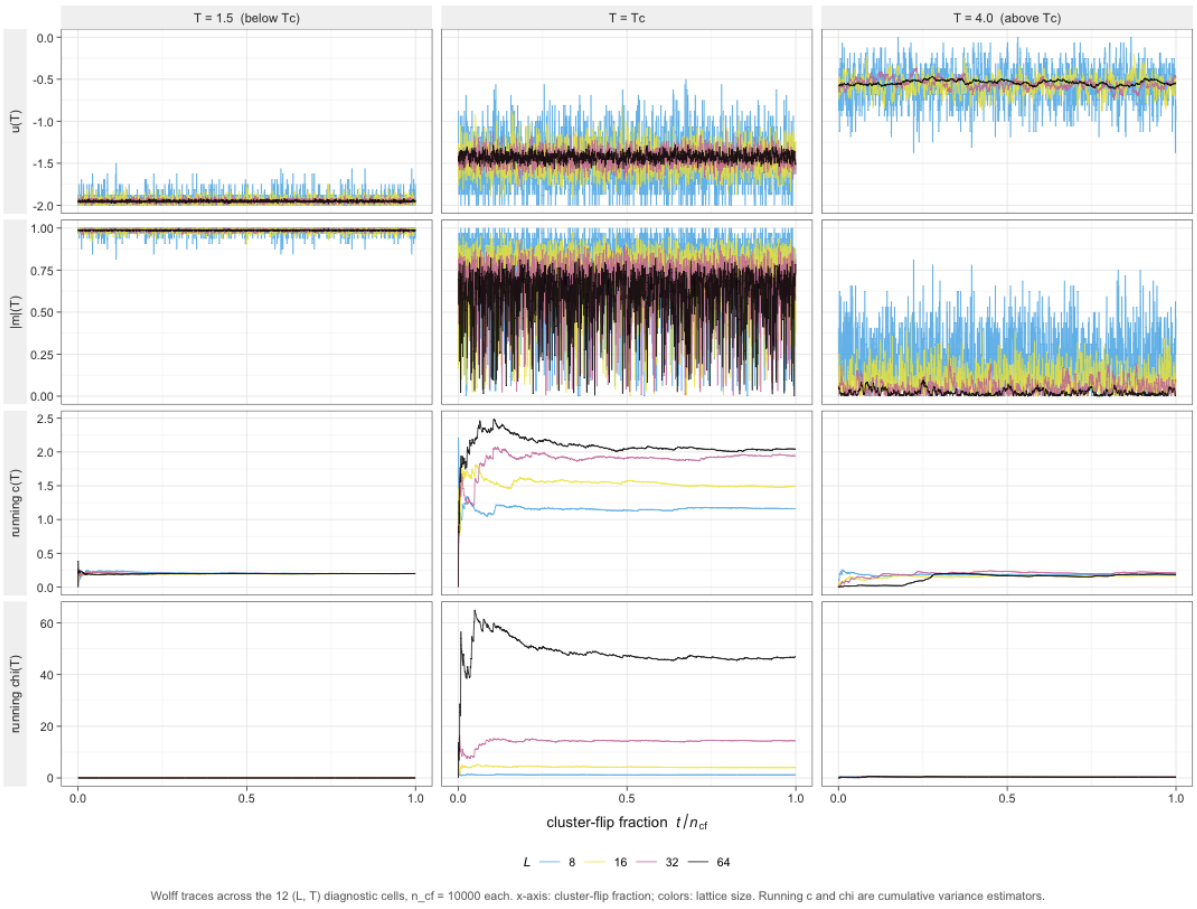


Figure 11. Wolff chain traces across the same diagnostic (L, T) grid as Figure 10, with the horizontal axis in cluster-flip fraction. All Wolff chains have the same length ($n_{\mathrm{cf}} = 10\,000$). Compared with Figure 10, the magnetization traces at $T = T_{\mathrm{c}}$ fluctuate around their stationary values from the start, with no visible drift even at $L = 64$.

B Sanity checks

Before any MCMC was run, every component of the analysis pipeline — lattice utilities, Onsager and Yang formulas, asymptotic limits, and the literature dynamic-exponent constants used as references — was verified against fourteen explicit numerical assertions, listed in Table 6. All fourteen pass. The lattice tests confirm that the Hamiltonian and incremental energy-change routines are consistent under periodic boundary conditions; the Onsager tests confirm that the elliptic-integral implementations reproduce $u(T_{\mathrm{c}}) = -\sqrt{2}\,J$ and the asymptotic specific-heat form near criticality; the Yang and high-temperature tests confirm the magnetization saturates at unity near $T = 0$ and tends to zero at large T; and the literature constants for $z_M^{\mathrm{Metropolis}}$ and z^{Wolff} are recorded for reference in Sections 5.1 and 5.2.

#	Assertion	Obtained	Expected	Status
1	Cold-lattice energy $L = 8$	-128	-128	PASS
2	Cold-lattice magnetization $L = 8$	$+64$	$+64$	PASS
3	ΔE flipping interior spin	$+8$	$+8$	PASS
4	Incremental vs full E after one flip	-120	-120	PASS
5	ΔE flipping corner spin (PBC)	$+8$	$+8$	PASS
6	T_c from Onsager exact formula	2.269185	2.269185	PASS
7	$u(T_c - 10^{-3})$	-1.418223	$-\sqrt{2}$	PASS
8	$c(T_c - 5 \times 10^{-3})$ full vs asymptotic	2.7190	2.7184	PASS
9	$\lvert m\rvert(T = 1.0)$ Yang formula	0.999276	≈ 1	PASS
10	$\lvert m\rvert(T = 3.0)$ Yang formula	0	0	PASS
11	c asymptotic convergence as $T \to T_c$	5.12×10^{-6}	$< 2.35 \times 10^{-4}$	PASS
12	$u(T = 100)$ high-temperature limit	-0.0200	≈ 0	PASS
13	$z_M^{\text{Metropolis}}$ literature constant	2.1665	[3]	PASS
14	z^{Wolff} literature constant	0.25	[4]	PASS

Table 6. Numerical sanity checks executed before any MCMC simulation. All fourteen assertions pass to the tolerance specified in the test. Assertions 1–5 test the lattice utilities; 6–12 test the Onsager/Yang exact-solution implementations and the asymptotic forms near criticality and at high temperature; 13–14 record the literature dynamic-exponent values used as references in Section 5.2.

C Implementation notes

Both samplers were implemented twice: a transparent pure-R reference following Sections 4.1.1 and 4.1.2 line by line, and an Rcpp inner-loop version for production runs. Both versions share R's random number stream via `GetRNGstate` and `PutRNGstate`, so a fixed R-level seed reproduces every chain bit-for-bit across sessions.

Three implementation choices in the Rcpp versions deserve brief discussion. First, the Metropolis sampler exploits the discreteness of $\Delta E \in \{-8J, -4J, 0, +4J, +8J\}$ on the square lattice: at the start of each run a five-entry lookup table of $e^{-\beta\Delta E}$ values is precomputed, and the accept/reject step indexes into this table by $(\Delta E + 8)/4$ rather than calling the transcendental `exp` routine in the inner loop. This optimization, combined with the move from R to C++, produces the median 341× speedup over pure R. Table 7 reports the per-cell speedups across the diagnostic grid; the floor of 64× at $L = 8, T = T_c$ comes from R-level overhead dominating the small lattice, and the ceiling of 578× at $L = 64, T = 4$ comes from the inner loop being almost entirely contiguous integer arithmetic at high temperature.

Second, the Wolff cluster construction uses an explicit last-in-first-out stack rather than the breadth-first queue often implied by "recursive cluster growth" in textbook presentations. The stack is allocated once at the start of each run as a flat integer vector of length N, and an in-cluster boolean flag array is zeroed before each cluster build. Same-sign neighbors of the spin popped from the top of the stack are tested for inclusion (with bond probability $p_{\text{add}} = 1 - e^{-2\beta J}$) and pushed onto the stack if accepted. The traversal terminates when the stack is empty; the resulting set of in-cluster sites is then flipped atomically. This produces the same distribution over clusters as a queue-based traversal — only the order in which boundary bonds are tested changes, and that order is irrelevant by detailed balance.

Third, after each Wolff cluster flip the energy and magnetization are recomputed from scratch in $O(N)$ time rather than maintained incrementally. Maintaining them incrementally would require iterating over the cluster boundary (whose size depends on cluster geometry, not just cluster size), and the bookkeeping is delicate enough to be error-prone for modest gain. The full recomputation amortizes well in practice: the typical cluster at $T = T_c$, $L = 64$ touches $|\bar{C}| = 1573$ spins, so a 4096-spin recomputation is cheap relative to

the cluster build itself.

L	regime	T	t_{R} [s]	t_{Rcpp} [s]	speedup
8	below T_c	1.500	0.677	0.003	226
8	at T_c	2.269	0.575	0.009	64
8	above T_c	4.000	0.570	0.002	285
16	below T_c	1.500	2.244	0.007	321
16	at T_c	2.269	2.267	0.007	324
16	above T_c	4.000	2.215	0.007	316
32	below T_c	1.500	8.783	0.024	366
32	at T_c	2.269	9.335	0.026	359
32	above T_c	4.000	10.737	0.030	358
64	below T_c	1.500	37.246	0.094	396
64	at T_c	2.269	48.010	0.102	471
64	above T_c	4.000	65.283	0.113	578

Table 7. Per-cell Metropolis speed comparison: pure R versus Rcpp inner loop, both seeded with `seed = 1` and run for $n_{\mathrm{swp}} = 500$. Median speedup 341×, range 64–578×. The $L = 8, T = T_c$ row has the smallest speedup because R-level per-call overhead dominates at small lattices. This table depends on CPU speed and can change with every run.

Figures 12 and 13 display the same data graphically along with mean cluster sizes for Wolff. The cluster size at T_c grows as $|\bar{C}| \sim L^{7/4}$, consistent with the Fortuin–Kasteleyn correspondence between bond clusters in the random-cluster model and the magnetization channel of the Ising model.

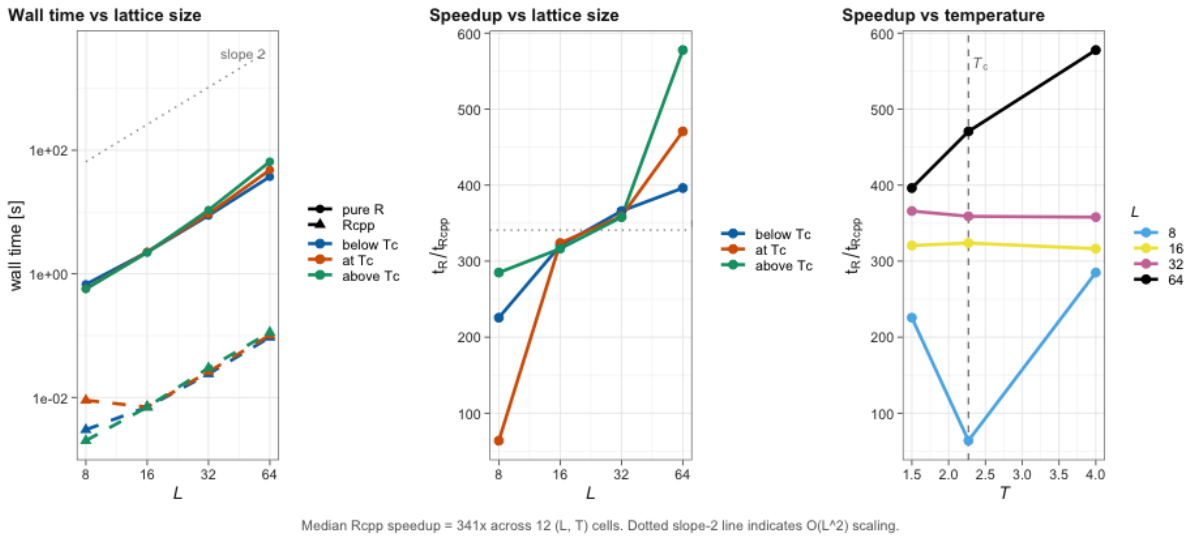


Figure 12. Pure-R vs Rcpp Metropolis benchmark across the diagnostic (L, T) grid. Left: wall time vs L on log-log axes, with both implementations and three temperature regimes overlaid; the dotted reference line has slope 2 (the expected $O(L^2)$ sweep cost). Middle: per-cell speedup vs L, with median across the grid annotated. Right: speedup vs T, with T_c marked. This figure depends on CPU speed and can change with every run.

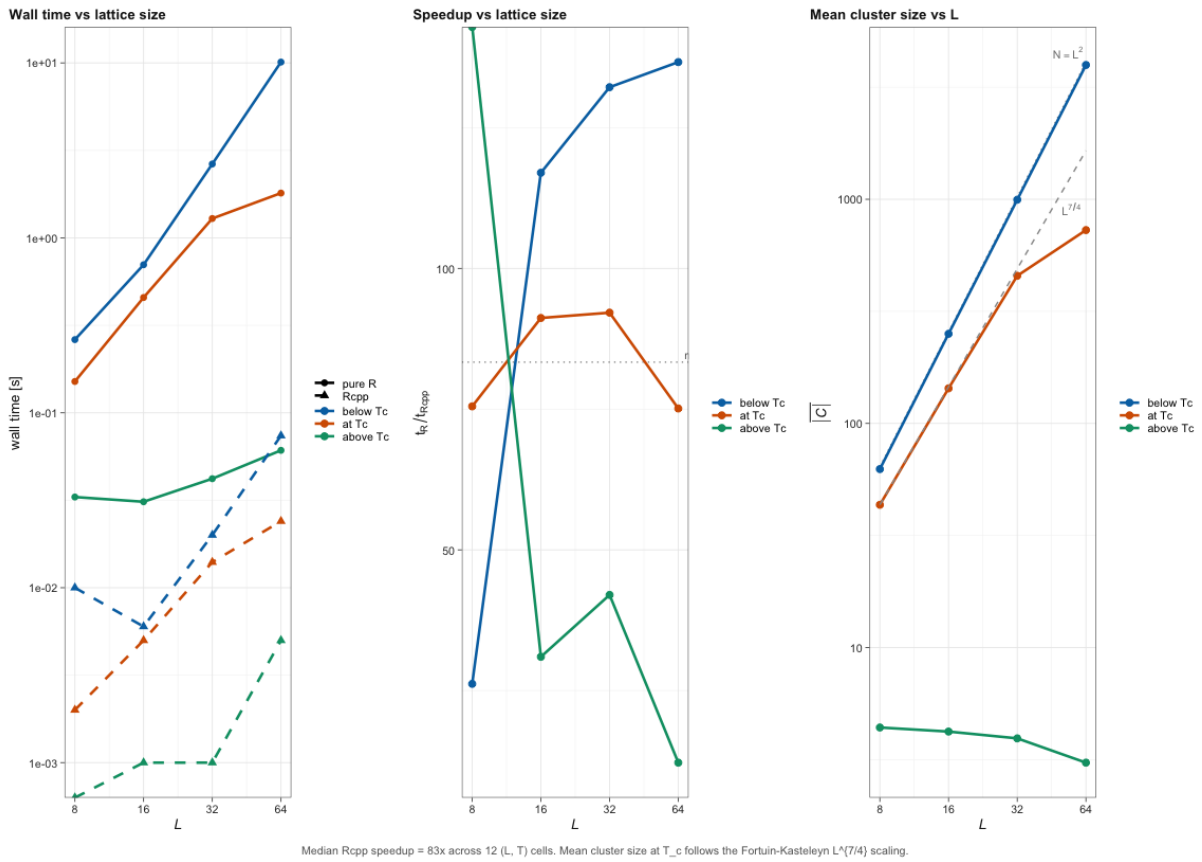


Figure 13. Pure-R vs Rcpp Wolff benchmark across the diagnostic (L, T) grid. The third panel shows mean cluster size $|\bar{C}|$ vs L across T-regimes; at T_c the growth is consistent with $L^{7/4}$ (dashed reference), and below T_c the cluster size approaches the upper bound $N = L^2$ (dotted lines per L). This figure depends on CPU speed and can change with every run.

The full pipeline reproduces from a single `set.seed` call in approximately 20 minutes on a single Apple M1 core (R 4.5, no parallelization). The dominant cost is the Metropolis $L = 64, T = T_c$ chain at 320 000 sweeps, which alone takes ~7 minutes; all remaining Metropolis cells, all Wolff cells, the full T-sweep with bootstrap SEs, and the bootstrap T_c extrapolation together take comparable total time.

D Block-bootstrap SE across observables

Sections 5.4 to 5.6 focus on the energy observable for the calibration analysis and on the $L = 32, T = T_c$ cell for the head-to-head comparison. Figure 14 reports block-bootstrap standard errors for all four observables across the full 34-point Wolff T-sweep underlying Figure 5, showing that the SE peaks at T_c are universal across observables and grow with L, mirroring the corresponding peaks in the point estimates.

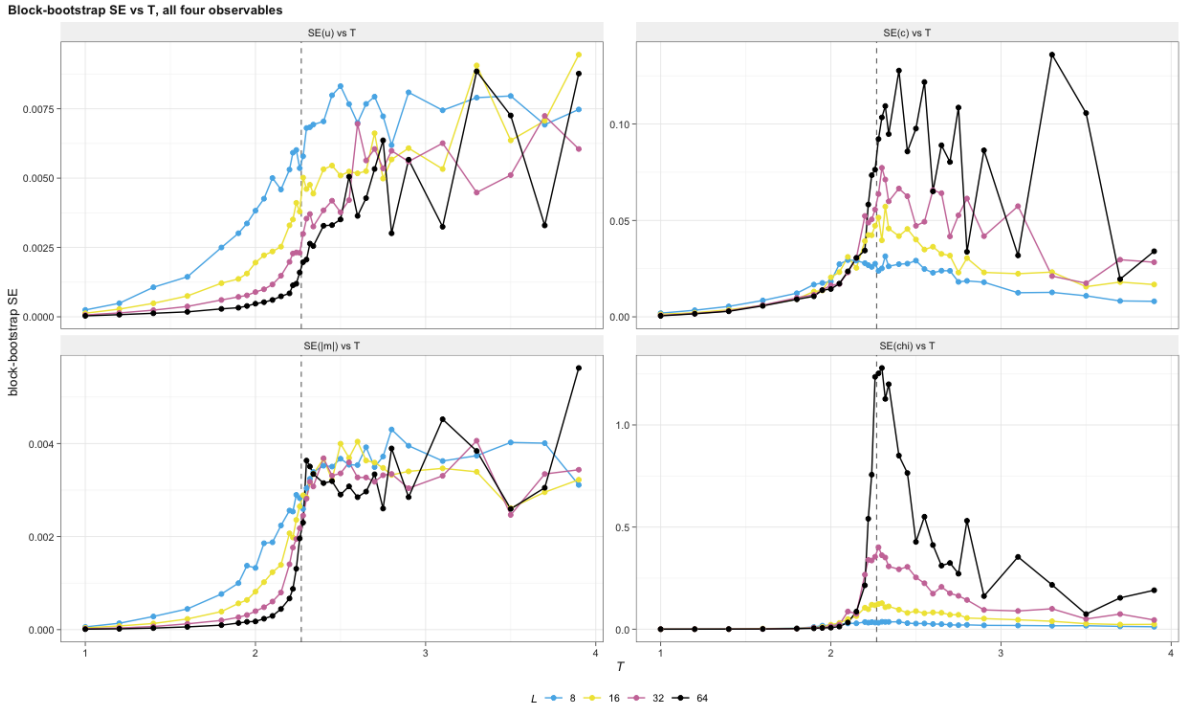


Figure 14. Block-bootstrap standard error vs temperature for the four observables, all four lattice sizes (Wolff). Each panel shows $\widehat{\mathrm{SE}}$ vs T for one observable: u, c, $|m|$, χ. The variance-based observables c and χ have SE peaks near T_{c} that grow with L, mirroring the corresponding peaks in the point estimates of Figure 5.